POLAR: Fractional Spectral Link Recommendation via Pole-Shared Krylov Sketching

(Extended Version with Complete Proofs)

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***Abstract*—Budgeted link insertion must balance local repairs and connections across weakly coupled regions. We study fractional inverse-trace reduction, a tunable spectral objective whose exponent changes the weighting of diffusion times. Optimizing this non-submodular family requires accurate fractional marginals and efficient repeated selection. POLAR combines positive Stieltjes quadrature, conditional greedy guarantees, and pole-shared Krylov solves with maintained endpoint sketches and algebraic re-ranking. As an optimizer, POLAR closely tracks exact fractional greedy on real networks, whereas the tested stochastic Lanczos quadrature configuration takes 12–$39\times$ as long with lower target gains. A four-million-vertex graph finishes in 15.1 minutes with shortlist size $B = 1$, or 28.7 minutes at the default $B = 16$. Whether the objective matters depends on the regularization σ, the shift added to the Laplacian. Under strong regularization ($\sigma = 1$) on three uniform candidate pools, inverse-trace and log-determinant selections lie within 2.5% of exact fractional greedy in target value, yet select different edges. Under weak regularization ($\sigma = 10^{-2}$) on a controlled graph where local and multiscale repairs compete, fractional target gains exceed those of inverse-trace selection by 22–48%.**

***Index Terms*—Spectral graph optimization, link recommendation, fractional matrix functions, rational quadrature, Krylov methods, non-submodular optimization.**

I. Introduction

An analyst adding links to a collaboration or communication graph may need to connect weakly coupled groups, reinforce peripheral vertices, or improve local connectivity. These repairs compete under an edge budget. Their value depends on the query of interest, such as current-flow group measures [1], [2] or forest-matrix proximity [3]. The task is to improve an existing graph, not predict future links.

Conventional criteria impose different preferences. Degree-based selection emphasizes local structure, whereas resistance and inverse-trace criteria emphasize global spectral responses [4]–[6]. A criterion suited to one diffusion horizon need not suit another. Fractional inverse-trace reduction provides a continuous preference: for an inserted edge set S with graph Laplacian L_S, it reduces $\mathrm{Tr}((L_S + \sigma I)^{-s})$, where Tr is the trace and I the identity. The exponent $0 < s < 1$ tunes the weighting of diffusion times, and larger s weights longer times more strongly at fixed σ. The regularization $\sigma > 0$ shifts the Laplacian, so disconnected graphs are admissible. Section II defines the resulting gain f_s and its log-determinant and inverse-trace endpoints. We seek an efficient optimizer and a characterization of when objective choice matters.

Existing scalable optimizers exploit either the single-resolvent rank-one update available at $s = 1$ [5] or submodularity [9], [36]; fractional negative powers offer neither. Three difficulties follow. Fractional powers lack a single-resolvent rank-one update, while subtracting approximate traces can lose relative marginal accuracy. Marginal returns can increase after an insertion, invalidating conventional lazy bounds (Fig. 1(a)). Finally, repeated shifted solves and sketch reconstruction are costly across thousands of candidates and multiple insertion rounds.

POLAR (Pole-shared Orthogonalized Lazy Algebraic Re-ranking) addresses these difficulties through Algorithm 1:

1) **Fractional marginal evaluation.** Positive Stieltjes quadrature gives dimension-free relative marginal accuracy.
2) **Non-submodular selection.** Ratio–curvature bounds yield a conditional error-aware greedy guarantee [7]. A lossless proxy-shortlist scheme supports a CPU-oriented lazy backend without diminishing returns; the GPU backend directly sorts scores.
3) **Cross-round reuse.** Shared Krylov iterations [8], endpoint sketches, norm caches, and algebraic re-ranking reduce k rounds to at most $2d_{\mathrm{sk}} + kB$ multi-shift right-hand sides, versus $2kd_{\mathrm{sk}}$ for rebuilding. Here d_{sk} is sketch dimension and B shortlist size.

Across eighteen real networks, POLAR matches tested exhaustive optima and attains 99.2–100% of exact fractional greedy on five medium networks; a greedy baseline based on stochastic Lanczos quadrature (SLQ) takes 12–$39\times$ as long with lower gains. No-Reuse, an ablation that rebuilds the sketches in every round, tests the savings predicted by the solve count. On uniform pools at $\sigma = 1$, exact $s = 0.25$ and inverse-trace selections have Jaccard similarity 0.43–0.74 despite target gains within 2.5%. When local and multiscale repairs compete at $\sigma = 10^{-2}$, POLAR exceeds inverse-trace

selection by 21.77–48.06% on the fractional targets (Fig. 1(b)).

II. Preliminaries, Problem Statement, and Spectral Motivation

A. Graph Model and Optimization Objectives

Let $G_0 = (V, E_0, w_0)$ be an undirected graph with $n = |V|$ vertices, nonnegative edge weights w_0, Laplacian L_0, and maximum weighted degree $d_{\max,0}$. Connectivity is not required. The candidate pool $\mathcal{C}$ contains $M = |\mathcal{C}|$ distinct non-loop edges absent from E_0; the insertion budget is $1 \le k \le M$. Each selected edge has unit weight. For $e = \{u, v\}$, let $\boldsymbol{e}_u, \boldsymbol{e}_v \in \mathbb{R}^n$ be the corresponding standard basis vectors and choose either orientation for $b_e = \boldsymbol{e}_u - \boldsymbol{e}_v$. Then $\|b_e\|_2^2 = 2$, and $b_e b_e^\top$ is independent of the orientation. Fix a regularization parameter $\sigma > 0$ and let I be the $n \times n$ identity. For any selected set $S \subseteq \mathcal{C}$, the added Laplacian ΔL_S, full Laplacian L_S, and regularized graph matrix are

$$\begin{aligned} \Delta L_S &= \sum_{e \in S} b_e b_e^\top, \qquad L_S = L_0 + \Delta L_S, \\ \mathsf{A}_S = \mathsf{A}(S) &= L_S + \sigma I, \qquad \sigma > 0. \end{aligned} \tag{1}$$

We write $\mathsf{A}_0 = \mathsf{A}_\varnothing$. Since $\mathsf{A}_S \succeq \sigma I \succ 0$, its fractional powers and logarithm are well-defined by spectral calculus, even for disconnected graphs. With Tr denoting the trace, the fractional gain and the log-determinant reference objective are

$$f_s(S) = \operatorname{Tr}(\mathsf{A}_0^{-s}) - \operatorname{Tr}(\mathsf{A}_S^{-s}), \quad 0 < s < 1, \tag{2}$$

$$\begin{aligned} f_{\log}(S) &= \operatorname{Tr}(\log \mathsf{A}_S) - \operatorname{Tr}(\log \mathsf{A}_0) \\ &= \log\det \mathsf{A}_S - \log\det \mathsf{A}_0. \end{aligned} \tag{3}$$

The shifted normalization preserves the reported gain units. In FastFrac's convention [9], the normalized matrix is $A_S^{\mathrm{FF}} = I + \alpha_{\mathrm{FF}} L_S$ with $\alpha_{\mathrm{FF}} = \sigma^{-1}$, so $\mathsf{A}_S = \sigma A_S^{\mathrm{FF}}$. Our gain is therefore $\sigma^{-s}[\operatorname{Tr}((A_0^{\mathrm{FF}})^{-s}) - \operatorname{Tr}((A_S^{\mathrm{FF}})^{-s})]$, whereas FastFrac uses positive powers. The objectives remain distinct.

The gains f_s and $f_{\log}$ vanish at $S = \varnothing$. The link recommendation problem is to maximize the fractional gain:

$$\max_{S \subseteq \mathcal{C},\ |S| \le k} f_s(S). \tag{4}$$

For $e \in \mathcal{C} \setminus S$, write $g_s(e \mid S)$ for its positive marginal gain and $\Delta_s(e; \mathsf{A}_S) = -g_s(e \mid S)$ for the signed trace change used in the algebraic formulas below.

$$g_s(e \mid S) = f_s(S \cup \{e\}) - f_s(S), \tag{5}$$

$$\begin{aligned} \Delta_s(e; \mathsf{A}_S) &= \operatorname{Tr}((\mathsf{A}_S + b_e b_e^\top)^{-s}) - \operatorname{Tr}(\mathsf{A}_S^{-s}) \\ &= -g_s(e \mid S). \end{aligned} \tag{6}$$

Let O be an optimal budget-k set. Comparisons with the greedy solution involve unions of at most $2k$ additions. Thus, for $|S| \le 2k$, the following spectral interval holds, where $A \preceq B$ denotes the positive-semidefinite order and κ_* bounds the condition number:

$$\sigma I \preceq \mathsf{A}_S \preceq \Lambda I, \qquad \Lambda = \sigma + 2d_{\max,0} + 4k, \qquad \kappa_* = \Lambda/\sigma. \tag{7}$$

The bounds follow, for any $x \in \mathbb{R}^n$, from $x^\top L_0 x \le 2d_{\max,0}\|x\|_2^2$ and $\|\Delta L_S\|_2 \le 2|S|$.

Later sketch updates only require coordinates at candidate endpoints. Let $V_{\mathcal{C}} = \bigcup_{e \in \mathcal{C}} \operatorname{supp}(b_e)$ be the set of vertices incident to candidate edges, where $\operatorname{supp}(b_e)$ denotes the nonzero coordinates of b_e, and let $n_{\mathcal{C}} = |V_{\mathcal{C}}| \le \min\{n, 2M\}$. For the algorithm, q is the number of quadrature poles, d_{sk} the sketch dimension, and B the re-ranked shortlist size. In complexity bounds, h denotes the maximum number of conjugate-gradient (CG) iterations per right-hand side.

B. Spectral Sensitivity and Budget Allocation

To quantify what a candidate contributes, interpolate between A_S and its single-edge update. For $e \in \mathcal{C} \setminus S$, let $\mathsf{A}_u = \mathsf{A}_S + u b_e b_e^\top$ for $0 \le u \le 1$. Here u is an edge-insertion parameter. Let $\lambda_i(u)$ and $v_i(u)$, for $1 \le i \le n$, be eigenvalues and orthonormal eigenvectors of $L_S + u b_e b_e^\top$. Thus A_u has eigenvalues $\lambda_i(u) + \sigma$. The following identity measures the full marginal by sensitivity along this update path.

Lemma 1 (Spectral sensitivity)**.** *For every $0 < s < 1$,*

$$\begin{aligned} g_s(e \mid S) &= s \int_0^1 b_e^\top \mathsf{A}_u^{-s-1} b_e \, du \\ &= \int_0^1 \sum_{i=1}^n \frac{s (v_i(u)^\top b_e)^2}{(\lambda_i(u) + \sigma)^{s+1}} \, du. \end{aligned} \tag{8}$$

In particular, every candidate has a strictly positive marginal.

Proof. Set $c_s = \sin(\pi s)/\pi > 0$. The Stieltjes identity, derived explicitly in (12), gives

$$\operatorname{Tr}(\mathsf{A}_u^{-s}) = c_s \int_0^\infty t^{-s} \operatorname{Tr}((\mathsf{A}_u + tI)^{-1}) \, dt,$$

Differentiating an inverse and using cyclic invariance of trace gives

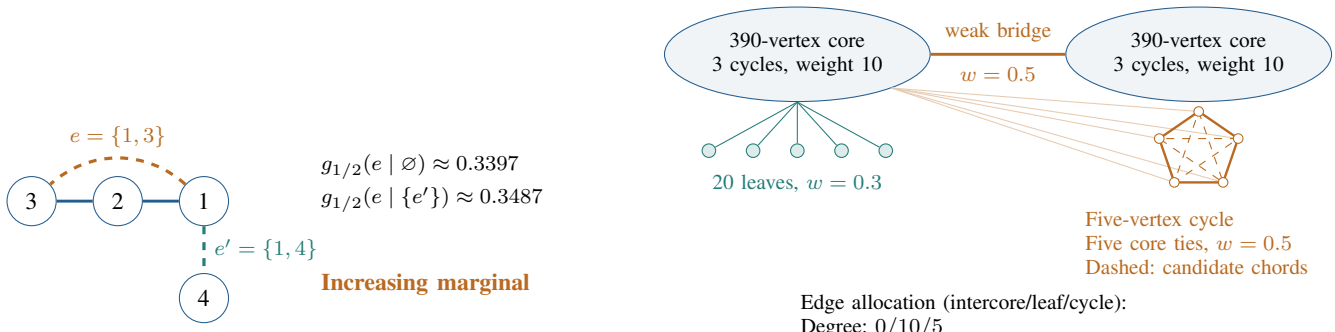
$$\frac{d}{du} \operatorname{Tr}((\mathsf{A}_u + tI)^{-1}) = -b_e^\top (\mathsf{A}_u + tI)^{-2} b_e.$$

The absolute value is at most $2/(\sigma + t)^2$. Its product with t^{-s} is integrable at zero because $s < 1$ and at infinity because $s > 0$. Differentiation under the integral is therefore valid. Differentiating the scalar Stieltjes identity and applying it to each eigenvalue of A_u yields

$$\begin{aligned} -\frac{d}{du} \operatorname{Tr}(\mathsf{A}_u^{-s}) &= c_s \int_0^\infty t^{-s} b_e^\top (\mathsf{A}_u + tI)^{-2} b_e \, dt \\ &= s b_e^\top \mathsf{A}_u^{-s-1} b_e. \end{aligned}$$

Integrating from zero to one proves the first equality in (8). Diagonalizing A_u proves the second. The quadratic form is positive because $\mathsf{A}_u^{-s-1} \succ 0$ and $b_e \neq 0$, proving strict positivity. □

For a simple eigenvalue, $\dot{\lambda}_i(u) = (v_i(u)^\top b_e)^2$, where the dot denotes differentiation along the insertion path. Its sensitivity weight is $w_s(\lambda) = s(\lambda + \sigma)^{-s-1}$. Smaller s reduces the relative dominance of low-frequency modes, a preference that also has a diffusion-time interpretation.



(a) A four-vertex counterexample (Example 1).

(b) Controlled multiscale graph ($n = 805$).

Fig. 1. Two aspects of fractional spectral link selection. (a) A four-vertex violation of diminishing returns: inserting e' increases the marginal gain of e. Solid edges belong to the base graph; dashed edges are candidates. (b) Schematic of the controlled multiscale benchmark ($n = 805$), comprising two dense cores, a weak bridge, peripheral leaves, and a locally anchored cycle. Five of the twenty leaves are illustrated. Edge allocations and fractional gains are reported in Table V.

Diffusion time scales: For diffusion time $\tau > 0$ and a Laplacian L, the heat trace $\operatorname{Tr}(e^{-\tau L})$ is the sum of continuous-time random-walk return probabilities. Let $\Delta\mathcal{H}_S(\tau) = \operatorname{Tr}(e^{-\tau L_0}) - \operatorname{Tr}(e^{-\tau L_S}) \geq 0$ denote its reduction after inserting S, and let $\Gamma(s)$ be the Gamma function. For the short-time expansion, define $\Delta_S \operatorname{Tr}(L^2) = \operatorname{Tr}(L_S^2) - \operatorname{Tr}(L_0^2)$. When inserting a candidate edge $\{u, v\}$, let d_u, d_v denote its endpoints' current weighted degrees. We write w_{uv} for the current edge weight between u and v (zero if absent).

Proposition 1 (Heat-trace representation)**.** *For $0 < s < 1$,*

$$f_s(S) = \frac{1}{\Gamma(s)} \int_0^\infty \tau^{s-1} e^{-\sigma\tau} \Delta\mathcal{H}_S(\tau)\, d\tau, \qquad (9)$$

$$f_{\log}(S) = \int_0^\infty \tau^{-1} e^{-\sigma\tau} \Delta\mathcal{H}_S(\tau)\, d\tau,$$

and the same formula with $s = 1$ gives the inverse-trace gain. As $\tau \to 0$, $\Delta\mathcal{H}_S(\tau) = 2|S|\tau - \frac{\tau^2}{2}\Delta_S \operatorname{Tr}(L^2) + O(\tau^3)$, where adding $\{u, v\}$ increases $\operatorname{Tr}(L^2)$ by $2(d_u + d_v) + 4$.

Proof. The fractional-power and logarithmic integral representations applied to the eigenvalues of A_0 and A_S, together with $\operatorname{Tr}(e^{-\tau\mathsf{A}}) = e^{-\sigma\tau} \operatorname{Tr}(e^{-\tau L})$ for $\mathsf{A} = L + \sigma I$, give the stated formulas. Monotonicity of the heat trace under Laplacian addition gives $\Delta\mathcal{H}_S(\tau) \geq 0$. Expanding the exponential and using $\operatorname{Tr}(L^2) = \sum_v d_v^2 + \sum_{u \neq v} w_{uv}^2$ yields the short-time terms. □

The kernel in (9) is proportional to a Gamma density with shape s, rate σ, and mean s/σ; consequently, f_s equals σ^{-s} times the corresponding expected heat-trace reduction. At fixed budget, selections share the leading short-time term; the next term favors smaller endpoint-degree sums. Increasing s emphasizes longer times, whereas increasing σ emphasizes shorter times.

Changing s can reverse edge rankings even as $\sigma \to 0^+$. When an insertion does not merge components, the zero-mode contributions cancel, leaving the difference of $\sum_{\lambda > 0} \lambda^{-s}$ over the nonzero Laplacian eigenvalues. Consider a unit-weight triangle on $\{1, 2, 3\}$, with leaves $4, 5$ attached to vertex 1 and leaf 6 attached to vertex 2. In the $\sigma \to 0^+$ limit, inserting $\{4, 5\}$ or $\{2, 4\}$ yields gains of 0.423 and 0.403 at $s = 1/2$, but $2/3$ and $13/18$ at $s = 1$, respectively. Their ranking therefore reverses. Indeed, the Laplacian characteristic polynomials of the base graph and of the graphs with $\{2, 4\}$ and $\{4, 5\}$ added are $x(x-1)Q(x)$, $x(x-2)(x^2-6x+4)(x^2-6x+6)$, and $x(x-3)Q(x)$, with $Q(x) = x^4 - 11x^3 + 39x^2 - 49x + 18$. Hence $\{4, 5\}$ gains $1 - 3^{-s}$ (it is a twin pair, cf. Proposition 5), the pseudoinverse traces are $67/18$, 3, and $55/18$, and the $s = \frac{1}{2}$ gain of $\{2, 4\}$ is 0.4025436208. Both strict orders persist for all sufficiently small $\sigma > 0$ by continuity. A link that joins two components instead adds $\sigma^{-s} + O(1)$ for every s, so with a candidate pool spanning components all exponents first favor merging links as $\sigma \to 0$.

C. An Exact Obstruction to Submodularity

Strictly positive marginals need not exhibit diminishing returns. For selected-edge sets $A \subseteq T \subseteq \mathcal{C}$, submodularity would require $g_s(e \mid A) \geq g_s(e \mid T)$ for any remaining edge $e \in \mathcal{C} \setminus T$: inserting other edges could never increase the value of a remaining edge. The following exact example violates this condition.

Example 1 (A four-vertex obstruction to submodularity)**.** Let $V = \{1, 2, 3, 4\}$, $E_0 = \{\{1, 2\}, \{2, 3\}\}$, $\sigma = 1/4$, and $s = 1/2$. All base and added edges have unit weight. Let $e = \{1, 3\}$ and $e' = \{1, 4\}$. For the normalized gain f_s, its marginals obey

$$g_s(e \mid \varnothing) = \frac{2}{\sqrt{5}} - \frac{2}{\sqrt{13}}, \qquad (10)$$

$$g_s(e \mid \{e'\}) = \frac{8\sqrt{2}}{7} + \frac{2}{3} - \frac{2}{\sqrt{5}} - \frac{2}{\sqrt{13}} - \frac{2}{\sqrt{17}}, \qquad (11)$$

and $g_s(e \mid \{e'\}) - g_s(e \mid \varnothing) > 1/375 > 0$. Thus f_s is not submodular, even for unweighted base graphs and unit-weight additions.

Proof. The base graph is a three-vertex path and an isolated vertex, so its Laplacian has eigenvalues $0, 0, 1, 3$. Indeed,

$(1,1,1,0)^\top$ and $(0,0,0,1)^\top$ are zero eigenvectors, while $(1,0,-1,0)^\top$ and $(1,-2,1,0)^\top$ have eigenvalues 1 and 3. Adding e gives a triangle and an isolated vertex, with eigenvalues $0,0,3,3$: every vector supported on the first three coordinates and summing to zero has eigenvalue 3.

Adding e' to the base graph gives the path 3–2–1–4, whose Laplacian is

$$L_{e'} = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}.$$

The subspaces $\{(x,x,y,y)^\top\}$ and $\{(x,-x,y,-y)^\top\}$ are invariant. On these subspaces, the coordinate matrices are respectively $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 3 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Their eigenvalues are $0, 2$ and $2-\sqrt{2}, 2+\sqrt{2}$, respectively. After adding both edges, the graph is a triangle with a pendant vertex. Its eigenvectors $(1,1,1,1)^\top$, $(0,1,1,-2)^\top$, $(0,1,-1,0)^\top$, and $(-3,1,1,1)^\top$ have eigenvalues $0,1,3,4$. These four independent vectors give the complete spectrum.

Adding σI shifts every eigenvalue by $1/4$. Subtracting the trace of the inverse square root after adding e from the trace before adding it therefore gives $g_s(e \mid \varnothing) = 2/\sqrt{5} - 2/\sqrt{13}$. For the second marginal, the three nonzero path eigenvalues contribute

$$\frac{2}{\sqrt{9-4\sqrt{2}}} + \frac{2}{3} + \frac{2}{\sqrt{9+4\sqrt{2}}}.$$

Both terms in the following sum are positive, and

$$\left(\frac{1}{\sqrt{9-4\sqrt{2}}} + \frac{1}{\sqrt{9+4\sqrt{2}}}\right)^2 = \frac{18}{49} + \frac{2}{7} = \frac{32}{49}.$$

Consequently the sum equals $4\sqrt{2}/7$. The common zero-eigenvalue contribution cancels, yielding the stated second marginal. Their difference is

$$D = \frac{8\sqrt{2}}{7} + \frac{2}{3} - \frac{4}{\sqrt{5}} - \frac{2}{\sqrt{17}}.$$

The inequalities $707^2 < 2 \cdot 500^2$, $16 \cdot 10000 < 5 \cdot 179^2$, and $4 \cdot 10000 < 17 \cdot 49^2$ imply

$$\sqrt{2} > \frac{707}{500}, \qquad \frac{4}{\sqrt{5}} < \frac{179}{100}, \qquad \frac{2}{\sqrt{17}} < \frac{49}{100}.$$

Substitution gives

$$D > \frac{8}{7}\frac{707}{500} + \frac{2}{3} - \frac{179}{100} - \frac{49}{100} = \frac{1}{375} > 0.$$

This strictly reverses the diminishing-returns inequality for $\varnothing \subset \{e'\}$ and $e \notin \{e'\}$. □

Numerically, the two marginals are 0.3397269948 and 0.3487121007; the strict inequality above is certified without floating-point arithmetic.

Hence stale marginal values cannot serve as valid lazy upper bounds; Sections III and IV provide non-submodular guarantees and safe proxy screening instead.

III. Quadrature-Based Marginal Formula and Approximation Guarantees

We derive a relative-error marginal oracle and non-submodular greedy bounds, then combine quadrature, proxy, and solve errors. This extended version gives complete proofs of all statements.

A. Positive Composite Quadrature

To avoid subtracting two approximate traces, we approximate each rank-one marginal through a positive Stieltjes representation. For $0 < s < 1$, set $c_s = \sin(\pi s)/\pi$; the inverse power satisfies the following identity for $x > 0$.

$$x^{-s} = c_s \int_0^\infty \frac{t^{-s}}{x+t}\, dt. \tag{12}$$

For a spectral interval $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$ with $0 < \lambda_{\mathrm{lo}} \le \lambda_{\mathrm{hi}}$, set $K = \lceil \log_2(\lambda_{\mathrm{hi}}/\lambda_{\mathrm{lo}}) \rceil$, $C = 2^K \lambda_{\mathrm{lo}}$, and $L_\ell = 2^\ell \lambda_{\mathrm{lo}}$. Partition the integral into $[0, \lambda_{\mathrm{lo}}]$, K middle intervals $[L_\ell, 2L_\ell]$, and $[C, \infty)$. Apply order-r Gauss–Jacobi rules with node–weight pairs (ℓ_i, A_i), (ξ_i, W_i), and (v_i, B_i) for the measures $u^{-s}du$ on $[0,1]$, du on $[-1,1]$, and $u^{s-1}du$ on $[0,1]$, respectively [10], [11]. The low, middle, and transformed high panels give the following positive poles and weights.

$$\begin{aligned} t_i^{\mathrm{lo}} &= \lambda_{\mathrm{lo}} \ell_i, & \omega_i^{\mathrm{lo}} &= c_s \lambda_{\mathrm{lo}}^{1-s} A_i, \\ t_{\ell i}^{\mathrm{mid}} &= \tfrac{L_\ell}{2}(3+\xi_i), & \omega_{\ell i}^{\mathrm{mid}} &= c_s \tfrac{L_\ell}{2} W_i (t_{\ell i}^{\mathrm{mid}})^{-s}, \\ t_i^{\mathrm{hi}} &= C/v_i, & \omega_i^{\mathrm{hi}} &= c_s C^{1-s} B_i / v_i. \end{aligned} \tag{13}$$

Enumerating these poles as $(t_j, \omega_j)_{j=1}^q$ gives $q = (K+2)r$ positive poles and weights, with neither tail truncated. Their sum defines the rational approximation below.

$$r_q(x) = \sum_{j=1}^q \frac{\omega_j}{x + t_j}. \tag{14}$$

B. Dimension-Free Relative Marginal Error

By Lemma 1, relative approximation of the positive derivative controls the marginal without subtracting approximate traces. Gaussian exactness provides this bound panelwise.

Lemma 2 (A positive Gaussian quadrature estimate)**.** *Let μ be a finite positive measure on $[-1,1]$ and let Q_r be a positive r-point Gaussian rule for μ. Suppose that F is analytic in a neighborhood of $\{z : |z| \le 2\}$, that $|F(z)| \le M_*$ on $|z| = 2$, and that $F(u) \ge m_* > 0$ on $[-1,1]$. Then*

$$\left| \int F\, d\mu - Q_r[F] \right| \le \frac{4M_*}{m_*} 4^{-r} \int F\, d\mu. \tag{15}$$

Moreover, for $\ell(u) = (1+u)/2$, provided $\int \ell\, d\mu > 0$,

$$\left| \int \ell F\, d\mu - Q_r[\ell F] \right| \le \frac{8M_*}{m_*} 4^{-r} \int \ell F\, d\mu. \tag{16}$$

Proof. Write the Taylor series $F(z) = \sum_{\nu=0}^\infty a_\nu z^\nu$. Cauchy's coefficient estimate on $|z| = 2$ gives $|a_\nu| \le M_* 2^{-\nu}$. For $P(u) = \sum_{\nu=0}^{2r-1} a_\nu u^\nu$ and $u \in [-1,1]$,

$$|F(u) - P(u)| \le M_* \sum_{\nu=2r}^\infty 2^{-\nu} = 2M_* 4^{-r}.$$

Exactness on P and on constants, together with positivity of the weights, yields

$$\begin{aligned}\left|\int F\,d\mu - Q_r[F]\right| &= \left|\int (F-P)\,d\mu - Q_r[F-P]\right| \\ &\le 2M_*4^{-r}\big(\mu([-1,1]) + Q_r[1]\big) \\ &= 4M_*4^{-r}\mu([-1,1]).\end{aligned}$$

Since $\int F\,d\mu \ge m_*\mu([-1,1])$, this proves (15).

For the second assertion, let $P_-(u) = \sum_{\nu=0}^{2r-2} a_\nu u^\nu$. Then

$$\sup_{[-1,1]} |F - P_-| \le M_* \sum_{\nu=2r-1}^{\infty} 2^{-\nu} = 4M_*4^{-r}.$$

The polynomial ℓP_- has degree at most $2r-1$. Since $\ell \ge 0$ and $Q_r[\ell] = \int \ell\,d\mu$, we obtain

$$\begin{aligned}\left|\int \ell F\,d\mu - Q_r[\ell F]\right| &\le 4M_*4^{-r}\left(\int \ell\,d\mu + Q_r[\ell]\right) \\ &= 8M_*4^{-r}\int \ell\,d\mu \\ &\le \frac{8M_*}{m_*}4^{-r}\int \ell F\,d\mu.\end{aligned}$$

This is (16). □

Lemma 3 (Uniform relative derivative approximation)**.** *For* (14)*, define*

$$D(x) = sx^{-s-1}, \qquad D_q(x) = \sum_{j=1}^{q} \frac{\omega_j}{(x+t_j)^2}.$$

For every $x \in [\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$,

$$|D_q(x) - D(x)| \le 256\,4^{-r} D(x). \tag{17}$$

Proof. Differentiating (12) under the integral gives

$$D(x) = c_s \int_0^\infty \frac{t^{-s}}{(x+t)^2}\,dt. \tag{18}$$

The integrable function $t^{-s}/(\lambda_{\mathrm{lo}}+t)^2$ justifies this differentiation uniformly for $x \in [\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$. We bound the relative error separately on each positive integral piece.

For the low tail, put $t = \lambda_{\mathrm{lo}} u$ and $v = 2u-1$. Apart from the positive factor $c_s {\lambda_{\mathrm{lo}}}^{1-s}(x+\lambda_{\mathrm{lo}}/2)^{-2}$, the kernel against $u^{-s}\,du$ is

$$F_{\mathrm{lo}}(v) = (1+\theta v)^{-2}, \qquad \theta = \frac{\lambda_{\mathrm{lo}}}{2x+\lambda_{\mathrm{lo}}} \le \frac{1}{3}.$$

On $|z| \le 2$, $|1+\theta z| \ge 1 - 2\theta \ge 1/3$, so the kernel is analytic on a neighborhood of this disk and has boundary modulus at most 9. For real $v \in [-1,1]$,

$$F_{\mathrm{lo}}(v) \ge (1+\theta)^{-2} \ge \frac{9}{16}.$$

Equation (15), applied to the affine image of the measure $u^{-s}\,du$, therefore gives the low-tail relative error $64\,4^{-r}$.

On a middle panel $[L, 2L]$, substitute $t = L(3+v)/2$. The positive prefactor is

$$c_s \frac{L}{2}\left(\frac{3L}{2}\right)^{-s}\left(x + \frac{3L}{2}\right)^{-2},$$

and the remaining kernel is

$$F_{\mathrm{mid}}(v) = (1+v/3)^{-s}(1+\theta v)^{-2}, \quad \theta = \frac{L}{2x+3L} \le \frac{1}{3}.$$

The branch positive on the real interval is analytic on $|z| \le 2$, because $\mathrm{Re}(1+z/3) \ge 1/3$. On its boundary,

$$|F_{\mathrm{mid}}(z)| \le 3^s 3^2 \le 27,$$

whereas on the real interval,

$$F_{\mathrm{mid}}(v) \ge (4/3)^{-s-2} \ge \frac{27}{64}.$$

Equation (15) gives relative error at most $256\,4^{-r}$ on every middle panel.

For the high tail, the substitution $t = C/u$ gives

$$c_s \int_C^\infty \frac{t^{-s}}{(x+t)^2}\,dt = c_s C^{1-s}\int_0^1 \frac{u^s}{(C+xu)^2}\,du.$$

The Gaussian rule in (13) is for $u^{s-1}\,du$. Its derivative integrand therefore contains an additional factor u. With $v = 2u-1$, the kernel, apart from $(C+x/2)^{-2}$, is

$$\ell(v)F_{\mathrm{hi}}(v), \quad \ell(v) = \frac{1+v}{2}, \quad F_{\mathrm{hi}}(v) = (1+\theta v)^{-2},$$

where $\theta = x/(2C+x) \le 1/3$, since $x \le \lambda_{\mathrm{hi}} \le C$. The same bounds $M_* = 9$ and $m_* = 9/16$ hold. Equation (16) gives the high-tail relative error $128\,4^{-r}$. This rule is precisely the high-tail contribution to D_q, because

$$\frac{\omega_i^{\mathrm{hi}}}{(x+t_i^{\mathrm{hi}})^2} = c_s C^{1-s} B_i \frac{v_i}{(C+xv_i)^2}.$$

Let $I_\nu(x) > 0$ be the exact derivative contribution of a panel, and let $\widetilde{I}_\nu(x)$ be its quadrature. Each preceding bound is at most $256\,4^{-r} I_\nu(x)$. Thus

$$\begin{aligned}|D_q(x) - D(x)| &\le \sum_\nu |\widetilde{I}_\nu(x) - I_\nu(x)| \\ &\le 256\,4^{-r}\sum_\nu I_\nu(x) = 256\,4^{-r}D(x).\end{aligned}$$

Positivity prevents a factor proportional to the number of panels from appearing. □

Spectral calculus extends the derivative bound to $\mathsf{A} + ub_eb_e^\top$, $0 \le u \le 1$, whenever its spectrum lies in $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$. Integration then gives a dimension-free relative marginal guarantee.

Theorem 1 (Dimension-free relative marginal approximation)**.** *Let* $\mathsf{A} \succ 0$ *and* $b_e \ne 0$*, and suppose that*

$$\mathrm{spec}(\mathsf{A} + ub_eb_e^\top) \subseteq [\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}] \quad \textit{for every } u \in [0,1].$$

Let $0 < s < 1$ *and* $0 < \epsilon \le 1/2$*. Construct* (14) *with* $r = \lceil \log_4(256/\epsilon)\rceil$*. All weights and poles are positive, and*

$$q = \big(\lceil \log_2(\lambda_{\mathrm{hi}}/\lambda_{\mathrm{lo}})\rceil + 2\big)r = O\big((1+\log(\lambda_{\mathrm{hi}}/\lambda_{\mathrm{lo}}))\log(1/\epsilon)\big).$$

The rational marginal is

$$\widetilde{\Delta}_s(e;\mathsf{A}) = -\sum_{j=1}^{q} \omega_j \frac{b_e^\top(\mathsf{A}+t_jI)^{-2}b_e}{1 + b_e^\top(\mathsf{A}+t_jI)^{-1}b_e}. \tag{19}$$

For $\Delta_s(e;\mathsf{A}) = \mathrm{Tr}((\mathsf{A}+b_e b_e^\top)^{-s}) - \mathrm{Tr}(\mathsf{A}^{-s})$*, it satisfies*

$$|\widetilde{\Delta}_s(e;\mathsf{A}) - \Delta_s(e;\mathsf{A})| \le \epsilon|\Delta_s(e;\mathsf{A})|. \tag{20}$$

Proof. Equation (13) gives positive weights and poles, numbering $(K+2)r$, which proves the pole-count assertion.

Write $R_j = (\mathsf{A}+t_jI)^{-1}$. Its positive definiteness implies $1 + b_e^\top R_j b_e > 0$. The Sherman–Morrison identity gives

$$(\mathsf{A}+t_jI+b_eb_e^\top)^{-1} = R_j - \frac{R_j b_e b_e^\top R_j}{1+b_e^\top R_j b_e}.$$

Taking traces and using their cyclic invariance yields

$$\mathrm{Tr}\big((\mathsf{A}+t_jI+b_eb_e^\top)^{-1} - R_j\big) = -\frac{b_e^\top R_j^2 b_e}{1+b_e^\top R_j b_e}.$$

Multiplication by ω_j and summation proves (19); it is exactly the trace difference of r_q.

Set $\mathsf{A}_u = \mathsf{A} + ub_eb_e^\top$. Differentiating the resolvent gives

$$\frac{d}{du}(\mathsf{A}_u+tI)^{-1} = -(\mathsf{A}_u+tI)^{-1}b_eb_e^\top(\mathsf{A}_u+tI)^{-1}.$$

The absolute value of its trace is at most $\|b_e\|^2/(\lambda_{\mathrm{lo}}+t)^2$. Since $\int_0^\infty t^{-s}/(\lambda_{\mathrm{lo}}+t)^2\,dt$ is finite, differentiation under the trace Stieltjes integral is justified. Consequently,

$$\begin{aligned}\frac{d}{du}\mathrm{Tr}(\mathsf{A}_u^{-s}) &= -c_s\int_0^\infty t^{-s}b_e^\top(\mathsf{A}_u+tI)^{-2}b_e\,dt\\ &= -b_e^\top D(\mathsf{A}_u)b_e.\end{aligned}$$

The last equality follows by applying (18) to each eigenvalue of A_u. Differentiating the finite rational sum similarly gives

$$\frac{d}{du}\mathrm{Tr}(r_q(\mathsf{A}_u)) = -b_e^\top D_q(\mathsf{A}_u)b_e.$$

The fundamental theorem of calculus now yields

$$-\Delta_s(e;\mathsf{A}) = \int_0^1 b_e^\top D(\mathsf{A}_u)b_e\,du,$$

and

$$-\widetilde{\Delta}_s(e;\mathsf{A}) = \int_0^1 b_e^\top D_q(\mathsf{A}_u)b_e\,du.$$

Lemma 3 and the chosen r imply

$$(1-\epsilon)D(x) \le D_q(x) \le (1+\epsilon)D(x) \quad (x\in[\lambda_{\mathrm{lo}},\lambda_{\mathrm{hi}}]).$$

Diagonalize $\mathsf{A}_u = V_u\Lambda_u V_u^\top$. Both $D(\mathsf{A}_u)$ and $D_q(\mathsf{A}_u)$ use this same eigenbasis, and every eigenvalue belongs to $[\lambda_{\mathrm{lo}},\lambda_{\mathrm{hi}}]$. Applying the scalar inequalities to each diagonal entry therefore proves

$$(1-\epsilon)D(\mathsf{A}_u) \preceq D_q(\mathsf{A}_u) \preceq (1+\epsilon)D(\mathsf{A}_u).$$

Taking quadratic forms with b_e and integrating gives

$$(1-\epsilon)(-\Delta_s(e;\mathsf{A})) \le -\widetilde{\Delta}_s(e;\mathsf{A}) \le (1+\epsilon)(-\Delta_s(e;\mathsf{A})).$$

Since $D(\mathsf{A}_u) = s\mathsf{A}_u^{-s-1} \succ 0$ and $b_e \neq 0$, the quantity $-\Delta_s(e;\mathsf{A})$ is positive. Rearranging proves (20). □

By (7), $\lambda_{\mathrm{lo}} = \sigma$ and $\lambda_{\mathrm{hi}} = \Lambda$ cover all relevant update paths of at most $2k$ edges.

For algebraic evaluation at state S, set $u_j = (\mathsf{A}_S + t_jI)^{-1}b_e$. By symmetry, the positive rational marginal is

$$G_q(e \mid S) = -\widetilde{\Delta}_s(e;\mathsf{A}_S) = \sum_{j=1}^q \omega_j \frac{\|u_j\|_2^2}{1+b_e^\top u_j}.$$

Theorem 1 controls G_q without projection error. To avoid evaluating every candidate algebraically, POLAR uses a proxy shortlist (Section IV).

C. The Log-Determinant Reference Objective

Proposition 2 (Strict monotonicity and submodularity of log-determinant gain)**.** *For the normalized log-determinant gain $f_{\log}$, every nonzero edge addition has positive marginal gain, and $f_{\log}$ is submodular. For two distinct candidates, diminishing returns is strict exactly when their inverse-matrix cross term is nonzero.*

Proof. Fix S and an unselected edge e, and write $R = \mathsf{A}_S^{-1}$. The matrix determinant lemma gives

$$g_{\log}(e \mid S) = \log(1 + b_e^\top R b_e) > 0,$$

because R is positive definite and $b_e \neq 0$. For $S \subseteq T$, let $A = \mathsf{A}_S$ and $H = \mathsf{A}_T - \mathsf{A}_S \succeq 0$. Writing $C = A^{-1/2}HA^{-1/2} \succeq 0$ gives

$$\mathsf{A}_T^{-1} = A^{-1/2}(I+C)^{-1}A^{-1/2} \preceq A^{-1},$$

since every eigenvalue of $(I+C)^{-1}$ is at most one. Thus $b_e^\top\mathsf{A}_T^{-1}b_e \le b_e^\top\mathsf{A}_S^{-1}b_e$; applying the increasing function $x \mapsto \log(1+x)$ proves diminishing returns.

More explicitly, for distinct unselected edges e, e', put $d_e = b_e^\top R b_e$, $d_{e'} = b_{e'}^\top R b_{e'}$, and $c = b_e^\top R b_{e'}$. Sherman–Morrison gives

$$\begin{aligned}&g_{\log}(e \mid S\cup\{e'\}) - g_{\log}(e \mid S)\\ &\quad= \log\left(1 - \frac{c^2}{(1+d_e)(1+d_{e'})}\right) \le 0.\end{aligned}$$

The logarithm is well-defined because Cauchy–Schwarz in the R-inner product yields $c^2 \le d_e d_{e'} < (1+d_e)(1+d_{e'})$. The inequality is strict exactly when $c \neq 0$.

For a connected equality example, take the unweighted four-cycle $E_0 = \{12, 23, 34, 14\}$ and its two missing diagonals $e = 13$ and $e' = 24$. Their incidence vectors are orthogonal eigenvectors of L_0, both with eigenvalue 2. Hence $Rb_{e'} = b_{e'}/(\sigma+2)$ and $b_e^\top R b_{e'} = 0$. The preceding equality shows that adding one diagonal does not change the log-determinant marginal of the other. □

D. Submodularity Ratio and Budget-Restricted Curvature

For disjoint $S, T \subseteq \mathcal{C}$, write $g_s(T \mid S) = f_s(S\cup T) - f_s(S)$. At budget $k \ge 1$, the restricted submodularity ratio compares individual and joint gains:

$$\gamma_k = \inf_{\substack{S\cap T=\varnothing,\ |S|\le k-1\\ 1\le|T|\le k}} \frac{\sum_{e\in T} g_s(e \mid S)}{g_s(T \mid S)}. \tag{21}$$

Restricted generalized curvature measures marginal deterioration across nested states:

$$\alpha_{2k} = 1 - \inf_{\substack{A \subseteq T,\ e \notin T \\ |T|+1 \le 2k}} \frac{g_s(e \mid T)}{g_s(e \mid A)}. \tag{22}$$

All sets are subsets of $\mathcal{C}$; Lemma 1 ensures positive denominators. Singleton T and $A = T$ give $\gamma_k \le 1$ and $\alpha_{2k} \ge 0$. The $2k$ restriction covers greedy–optimal unions.

Theorem 2 (Submodularity ratio and curvature)**.** *Let $0 < s < 1$, $\sigma > 0$, and unit-weight candidate edges with $\|b_e\|_2^2 = 2$. With Λ defined in* (7)*, f_s is normalized and strictly increasing, and the following bounds hold:*

$$\gamma_k \ge \gamma_0 := \frac{\sigma}{2s}\left[1 - \left(\frac{\sigma}{\sigma+2}\right)^s\right] > 0, \tag{23}$$

$$\alpha_{2k} \le \alpha_0 := 1 - \left(\frac{\sigma}{\Lambda}\right)^{s+1} < 1. \tag{24}$$

In fact, the bound in (23) *holds without a cardinality restriction on S or T.*

Proof. Set $c_s = \sin(\pi s)/\pi > 0$. Fix an SPD matrix $A \succeq \sigma I$ and a nonzero unit-edge vector b. For $t > 0$, write $R_t = (A + tI)^{-1}$, $d_t = b^\top R_t b$, and $\phi_t = b^\top R_t^2 b$. The Stieltjes representation and the Sherman–Morrison identity give

$$\operatorname{Tr}(A^{-s}) - \operatorname{Tr}((A + bb^\top)^{-s}) = c_s \int_0^\infty t^{-s} \frac{\phi_t}{1 + d_t}\, dt. \tag{25}$$

Every integrand is strictly positive. The integral is finite because it is $O(t^{-s})$ at zero and $O(t^{-s-2})$ at infinity. Consequently every candidate has a strictly positive marginal, and $f_s(\varnothing) = 0$ proves normalization and strict monotonicity.

Differentiating the scalar Stieltjes identity under the integral gives

$$s x^{-s-1} = c_s \int_0^\infty \frac{t^{-s}}{(x+t)^2}\, dt.$$

This differentiation is justified locally uniformly for $x > 0$ by an integrable bound of the form Ct^{-s} on $(0, 1)$ and Ct^{-s-2} on $(1, \infty)$. Applying the resulting identity in an eigenbasis of A yields

$$D(A, b) := s b^\top A^{-s-1} b = c_s \int_0^\infty t^{-s} \phi_t\, dt. \tag{26}$$

Since $d_t \le \|b\|_2^2/(\sigma + t) = 2/(\sigma + t)$,

$$\operatorname{Tr}(A^{-s}) - \operatorname{Tr}((A + bb^\top)^{-s}) \ge c_s \int_0^\infty t^{-s} \phi_t \frac{\sigma + t}{\sigma + t + 2}\, dt. \tag{27}$$

Let $\mu_i \ge \sigma$ be the eigenvalues of A, with orthonormal eigenvectors v_i, and let $z_i = (v_i^\top b)^2 \ge 0$. Then $\phi_t = \sum_i z_i/(\mu_i + t)^2$. For $\mu \ge \sigma$, define

$$I(\mu) = \int_0^\infty \frac{t^{-s}}{(\mu + t)^2} \frac{\sigma + t}{\sigma + t + 2}\, dt.$$

The substitution $t = \mu u$ gives

$$\begin{aligned} I(\mu) &= \mu^{-s-1} \int_0^\infty \frac{u^{-s}}{(1+u)^2} \frac{\sigma + \mu u}{\sigma + \mu u + 2}\, du \\ &\ge \left(\frac{\mu}{\sigma}\right)^{-s-1} I(\sigma). \end{aligned} \tag{28}$$

Indeed, $y/(y+2)$ is increasing for $y > 0$, and $\mu \ge \sigma$. At the endpoint $\mu = \sigma$, partial fractions and the Stieltjes identity imply

$$\begin{aligned} c_s I(\sigma) &= c_s \int_0^\infty \frac{t^{-s}}{(\sigma + t)(\sigma + t + 2)}\, dt \\ &= \frac{1}{2}\left[\sigma^{-s} - (\sigma + 2)^{-s}\right] = \gamma_0 s \sigma^{-s-1}. \end{aligned}$$

Substituting this equality into (28) and then into (27) proves

$$\operatorname{Tr}(A^{-s}) - \operatorname{Tr}((A + bb^\top)^{-s}) \ge \gamma_0 s \sum_i z_i \mu_i^{-s-1} = \gamma_0 D(A, b). \tag{29}$$

To compare this single-edge inequality with a joint gain, let $H = \sum_{e \in T} b_e b_e^\top \succeq 0$ and $K_t = R_t^{1/2} H R_t^{1/2} \succeq 0$. Since $x/(1+x) \le x$ for $x \ge 0$, the spectral theorem gives

$$0 \preceq I - (I + K_t)^{-1} \preceq K_t.$$

Congruence with $R_t^{1/2}$ therefore gives

$$\begin{aligned} R_t - (A + H + tI)^{-1} &= R_t^{1/2}[I - (I + K_t)^{-1}] R_t^{1/2} \\ &\preceq R_t H R_t. \end{aligned}$$

Taking traces, integrating with positive weight $c_s t^{-s}$, and using (26) yields

$$\operatorname{Tr}(A^{-s}) - \operatorname{Tr}((A + H)^{-s}) \le \sum_{e \in T} D(A, b_e). \tag{30}$$

Apply (29) to each $e \in T$ at $A = \mathsf{A}_S$ and sum. Combining the result with (30) proves $\sum_{e \in T} g_s(e \mid S) \ge \gamma_0 g_s(T \mid S)$. No cardinality restriction was used in this argument. The identity

$$\gamma_0 = \int_0^1 \left(\frac{\sigma}{\sigma + 2t}\right)^{s+1} dt$$

also shows directly that $0 < \gamma_0 < 1$.

It remains to prove the curvature bound. For every vector x,

$$\begin{aligned} x^\top L_0 x &= \sum_{\{u,v\} \in E_0} w_{uv}(x_u - x_v)^2 \\ &\le 2 \sum_v d_{v,0} x_v^2 \le 2 d_{\max,0} \|x\|_2^2. \end{aligned}$$

Each added edge has operator norm $\|b_e b_e^\top\|_2 = 2$. Thus every matrix formed from at most $2k$ added unit edges, including any partially added last edge, has all eigenvalues in $[\sigma, \Lambda]$. The resolvent derivative $\frac{d}{du}(A + ubb^\top + tI)^{-1} = -(A + ubb^\top + tI)^{-1} bb^\top (A + ubb^\top + tI)^{-1}$, the same integrable bounds as above, and cyclicity of the trace give

$$g_s(e \mid S) = s \int_0^1 b_e^\top (\mathsf{A}_S + u b_e b_e^\top)^{-s-1} b_e\, du. \tag{31}$$

The spectral theorem, applied separately at each u, implies

$$\Lambda^{-s-1} I \preceq (\mathsf{A}_S + u b_e b_e^\top)^{-s-1} \preceq \sigma^{-s-1} I.$$

Hence whenever $|S| + 1 \le 2k$,

$$2s\Lambda^{-s-1} \le g_s(e \mid S) \le 2s\sigma^{-s-1}.$$

For every pair $A \subseteq T$ appearing in (22), these bounds give

$$\frac{g_s(e \mid T)}{g_s(e \mid A)} \geq \left(\frac{\sigma}{\Lambda}\right)^{s+1}.$$

Taking the infimum proves (24). The comparison applies scalar spectral bounds separately to each path matrix. □

E. A Complete Approximate-Greedy Guarantee

Following [7], we analyze approximate greedy selection with a recovery factor β. Let S_i be the set after round i, e_i the selected edge, and O an optimal budget-k set.

Lemma 4 (Approximate greedy selection)**.** *Let f_s be normalized and monotone, with well-defined parameters $\gamma_k \geq \gamma > 0$ and $\alpha_{2k} \leq \alpha \leq 1$ from* (21)–(22)*. Let $S_0 = \varnothing$ and $S_i = S_{i-1} \cup \{e_i\}$, where*

$$g_s(e_i \mid S_{i-1}) \geq \beta \max_{e \in \mathcal{C} \setminus S_{i-1}} g_s(e \mid S_{i-1}), \qquad 0 < \beta \leq 1. \tag{32}$$

For an optimal set O of cardinality at most k, put $\theta = \beta\gamma$ and

$$\Phi_k(\alpha, \theta) := \frac{\theta}{k} \sum_{\ell=0}^{k-1} \left(1 - \frac{\alpha\theta}{k}\right)^{\ell}. \tag{33}$$

Then $f_s(S_k) \geq \Phi_k(\alpha, \theta) f_s(O)$. For $\alpha > 0$,

$$\Phi_k(\alpha, \theta) = \frac{1 - (1 - \alpha\theta/k)^k}{\alpha} \geq \frac{1 - e^{-\alpha\theta}}{\alpha};$$

for $\alpha = 0$, $\Phi_k(0, \theta) = \theta$.

Proof. If $f_s(O) = 0$ the claim is immediate; otherwise extend O to size k by monotonicity. Put $\rho_i = g_s(e_i \mid S_{i-1})$, $x_i = \rho_i / f_s(O)$, and $w_r = |S_r \cap O|$. Adding $S_r \setminus O$ to O in greedy order and applying curvature, then using the ratio and approximate-greedy conditions, gives

$$f_s(O \cup S_r) - f_s(O) \geq (1 - \alpha) \sum_{i \leq r : e_i \notin O} \rho_i,$$
$$f_s(O \cup S_r) - f_s(S_r) \leq (k - w_r)\rho_{r+1}/\theta.$$

Only unions of at most $2k$ edges occur. Combining yields

$$1 \leq \alpha \sum_{i \leq r : e_i \notin O} x_i + \sum_{i \leq r : e_i \in O} x_i + (k - w_r) x_{r+1}/\theta.$$

For the scalar relaxation, let $zC_{m,R}$ be the minimum remaining gain with residual $z \geq 0$, $m \leq R$ selections, and R unselected elements of O. Set $C_{0,R} = 0$. An inside increment $x = z$ clears the residual, so $0 \leq C_{m,R} \leq 1$. Therefore $x + C_{\text{next}} \max\{z - cx, 0\}$, with C_{next} the next-stage coefficient, is nondecreasing in x for $c \in \{\alpha, 1\}$, and the least admissible increment $x = \theta z / R$ minimizes either branch. Outside/inside choices reduce the residual by $\alpha x / x$, respectively; only the latter reduces R. Thus

$$C_{m,R} = \frac{\theta}{R} + \min\big\{(1 - \alpha\theta/R) C_{m-1,R}, \; (1 - \theta/R) C_{m-1,R-1}\big\}.$$

Induction gives $C_{m,R} = (\theta/R) \sum_{j=0}^{m-1} (1 - \alpha\theta/R)^j$, where $0 \leq \alpha\theta \leq 1$. The outside branch attains it; the inside branch is no smaller because, for $0 \leq j \leq m - 2$,

$$\frac{R - \theta}{R - 1}(1 - \alpha\theta/(R-1))^j \geq (1 - \alpha\theta/R)^{j+1}.$$

The case $R = 2$ is immediate; for $R \geq 3$ the ratio on the right to the power term on the left increases with j and is at most $J(R)/J(R-1) \leq 1$, where $J(u) = (1 - \alpha\theta/u)^{u-1}$. Indeed, $(\log J)'(u) \leq \log(1 - \alpha\theta/u) + \alpha\theta/u \leq 0$ for $u \geq 2$. At $(m, R, z) = (k, k, 1)$ this gives $\Phi_k(\alpha, \theta)$; the geometric sum and $(1 - \alpha\theta/k)^k \leq e^{-\alpha\theta}$ give the remaining assertions. □

We next relate β to the numerical errors of POLAR's marginal evaluations.

F. Numerical Errors and Algebraic Re-ranking

Algebraic-score accuracy additionally requires control of solve residuals.

Lemma 5 (Residual certification of algebraic reranking scores)**.** *Fix a state S, a candidate e, and positive quadrature weights ω_j. For each pole, let $A_j = \mathsf{A}_S + t_j I$, $u_j = A_j^{-1} b_e$, and $\lambda_{\text{lo,j}} I \preceq A_j \preceq \lambda_{\text{hi,j}} I$, with $\lambda_{\text{lo,j}} = \sigma + t_j$ and $\lambda_{\text{hi,j}} = \Lambda + t_j$. For some $0 < \rho < 1$, suppose the computed solution $\widetilde{u}_j$ satisfies the following residual condition.*

$$\|A_j \widetilde{u}_j - b_e\|_2 \leq \rho \frac{\lambda_{\text{lo,j}}}{\lambda_{\text{hi,j}}} \|b_e\|_2 \qquad (j = 1, \ldots, q). \tag{34}$$

Define $\check{G}(e \mid S)$ by replacing each u_j in the rational score $G_q(e \mid S)$ with $\widetilde{u}_j$. Then all computed denominators are positive, and the following interval holds.

$$\frac{(1-\rho)^2}{1+\rho} G_q(e \mid S) \leq \check{G}(e \mid S) \leq \frac{(1+\rho)^2}{1-\rho} G_q(e \mid S). \tag{35}$$

In particular, if $0 < \varepsilon_r \leq 1/2$ and $\rho \leq \varepsilon_r/4$, then $(1-\varepsilon_r) G_q \leq \check{G} \leq (1+\varepsilon_r) G_q$.

Proof. Write $r_j = A_j \widetilde{u}_j - b_e$ and $v_j = \widetilde{u}_j - u_j = A_j^{-1} r_j$. The assumed residual bound gives

$$\|v_j\|_2 \leq \lambda_{\text{lo,j}}^{-1} \|r_j\|_2 \leq \rho \frac{\|b_e\|_2}{\lambda_{\text{hi,j}}}.$$

Also, $\|b_e\|_2 = \|A_j u_j\|_2 \leq \lambda_{\text{hi,j}} \|u_j\|_2$ and $d_j := b_e^\top u_j \geq \|b_e\|_2^2 / \lambda_{\text{hi,j}} > 0$. Consequently

$$\|v_j\|_2 \leq \rho \|u_j\|_2, \qquad |b_e^\top v_j| \leq \|b_e\|_2 \|v_j\|_2 \leq \rho d_j.$$

The triangle and reverse triangle inequalities imply

$$(1-\rho)^2 \|u_j\|_2^2 \leq \|\widetilde{u}_j\|_2^2 \leq (1+\rho)^2 \|u_j\|_2^2.$$

For the denominator,

$$1 + b_e^\top \widetilde{u}_j \geq 1 + (1-\rho) d_j \geq (1-\rho)(1 + d_j) > 0,$$
$$1 + b_e^\top \widetilde{u}_j \leq 1 + (1+\rho) d_j \leq (1+\rho)(1 + d_j).$$

Combining these inequalities proves (35) for each summand; positivity of the weights preserves the bounds after summation. Finally, $\rho \leq \varepsilon_r/4 \leq 1/8$ implies

$$\frac{(1+\rho)^2}{1-\rho} - 1 = \frac{3\rho + \rho^2}{1-\rho} \leq 4\rho \leq \varepsilon_r,$$

where $\rho \le 1/5$ suffices for the middle inequality, and

$$1 - \frac{(1-\rho)^2}{1+\rho} = \frac{3\rho - \rho^2}{1+\rho} \le 3\rho \le \varepsilon_r.$$

These inequalities prove the last assertion. □

We combine quadrature, proxy, and re-ranking errors $\varepsilon_Q, \varepsilon_c, \varepsilon_r$. Proxy error includes both projection and numerical maintenance; the shortlist must retain a global proxy maximizer.

Theorem 3 (End-to-end guarantee with certified errors)**.** *Let G_q be the positive multi-pole marginal, and suppose quadrature satisfies, uniformly over all states of size less than k and their remaining candidates,*

$$(1-\varepsilon_Q) g_s(e \mid S) \le G_q(e \mid S) \le (1+\varepsilon_Q) g_s(e \mid S). \quad (36)$$

Suppose that on an event $\mathcal{E}$, the coarse proxy scores $\widehat{G}$ (Section III-G) satisfy

$$(1-\varepsilon_c) G_q(e \mid S) \le \widehat{G}(e \mid S) \le (1+\varepsilon_c) G_q(e \mid S) \quad (37)$$

for all candidates at every state actually visited. At each round, let the shortlist contain a global maximizer of $\widehat{G}$, and select from it a maximizer of a reranking score satisfying

$$(1-\varepsilon_r) G_q(e \mid S) \le \check{G}(e \mid S) \le (1+\varepsilon_r) G_q(e \mid S). \quad (38)$$

Assume $0 \le \varepsilon_Q, \varepsilon_c, \varepsilon_r < 1$, and put

$$\beta = \frac{1-\varepsilon_Q}{1+\varepsilon_Q} \frac{1-\varepsilon_c}{1+\varepsilon_c} \frac{1-\varepsilon_r}{1+\varepsilon_r}.$$

Then, on $\mathcal{E}$, the resulting set satisfies

$$f_s(S_k) \ge \Phi_k(\alpha_0, \beta\gamma_0) f_s(O) \ge \frac{1 - e^{-\alpha_0 \beta \gamma_0}}{\alpha_0} f_s(O), \quad (39)$$

where O is optimal under the budget k and γ_0, α_0 are given in (23)–(24)*. If $\Pr(\mathcal{E}) \ge 1-\delta$, the same guarantee holds with probability at least $1-\delta$.*

Proof. Work on $\mathcal{E}$ and fix a visited state S. Let c be a global maximizer of $\widehat{G}(\cdot \mid S)$ contained in the shortlist, let $e^\star$ be the reranking winner, and let e be any remaining candidate. Suppress S in the following display. Successive applications of the three error bounds and the two maximization rules give

$$\begin{aligned} g_s(e^\star) &\ge \frac{G_q(e^\star)}{1+\varepsilon_Q} \\ &\ge \frac{\check{G}(e^\star)}{(1+\varepsilon_Q)(1+\varepsilon_r)} \\ &\ge \frac{\check{G}(c)}{(1+\varepsilon_Q)(1+\varepsilon_r)} \\ &\ge \frac{(1-\varepsilon_r) G_q(c)}{(1+\varepsilon_Q)(1+\varepsilon_r)} \\ &\ge \frac{(1-\varepsilon_r)\widehat{G}(c)}{(1+\varepsilon_Q)(1+\varepsilon_r)(1+\varepsilon_c)} \\ &\ge \frac{(1-\varepsilon_r)\widehat{G}(e)}{(1+\varepsilon_Q)(1+\varepsilon_r)(1+\varepsilon_c)} \\ &\ge \frac{(1-\varepsilon_r)(1-\varepsilon_c) G_q(e)}{(1+\varepsilon_Q)(1+\varepsilon_r)(1+\varepsilon_c)} \\ &\ge \beta g_s(e). \end{aligned}$$

Thus the selected edge satisfies (32) at every round. Theorem 2 supplies the inequalities required by Lemma 4 with $\gamma = \gamma_0$ and $\alpha = \alpha_0$. Applying that lemma proves (39). The event implication immediately gives the probability statement. □

Remark 1 (Round-dependent recovery)**.** The factor β is a worst case over rounds. If round i satisfies (32) with its own β_i, a ratio-only argument gives $f_s(O) - f_s(S_i) \le (1 - \gamma_k \beta_i / k)(f_s(O) - f_s(S_{i-1}))$, hence $f_s(S_k) \ge \big(1 - e^{-\gamma_k \bar{\beta}}\big) f_s(O)$ with $\bar{\beta} = k^{-1} \sum_i \beta_i$.

G. A Uniform Certificate for Fixed Sketches

Fixed sketches reused adaptively require simultaneous rather than pointwise projection bounds [12], [13]. Here d_{sk} is the sketch dimension; η and δ control distortion and failure probability.

Corollary 1 (Uniform fixed-sketch guarantee)**.** *Fix $0 < \eta < 1/3$, $0 < \delta < 1$, $1 \le k \le M$, and positive quadrature weights and shifts $(\omega_j, t_j)_{j=1}^q$. Let m_0 be the number of base edges and $C_S \in \mathbb{R}^{(m_0+M)\times n}$ the weighted incidence factor comprising rows $\sqrt{w_a} b_a^\top$ for base edges $a \in E_0$ of weight w_a and incidence vector b_a, and $b_e^\top$ for selected candidates, with zero rows for unselected candidates. Thus $C_S^\top C_S = \mathsf{A}_S - \sigma I$. For $|S| < k$, $e \in \mathcal{C} \setminus S$, and $1 \le j \le q$, let $v_{S,e,j} = (\mathsf{A}_S + t_j I)^{-1} b_e$, $\phi_{S,e,j} = \|v_{S,e,j}\|^2$, $z_{S,e,j} = \|C_S v_{S,e,j}\|^2$, and $d_{S,e,j} = b_e^\top v_{S,e,j}$. Draw independent fixed Gaussian projections $Z \in \mathbb{R}^{n \times d_{\mathrm{sk}}}$ and $U \in \mathbb{R}^{(m_0+M)\times d_{\mathrm{sk}}}$, each with independent $\mathcal{N}(0, 1/d_{\mathrm{sk}})$ entries. Define $\widehat{\phi}_{S,e,j} = \|Z^\top v_{S,e,j}\|^2$ and $\widehat{z}_{S,e,j} = \|U^\top C_S v_{S,e,j}\|^2$, and form the proxy*

$$\widehat{G}(e \mid S) = \sum_{j=1}^q \omega_j \frac{\widehat{\phi}_{S,e,j}}{1 + \widehat{z}_{S,e,j} + (\sigma + t_j)\widehat{\phi}_{S,e,j}}. \quad (40)$$

Let $\mathcal{M}_k = \sum_{\ell=0}^{k-1} \binom{M}{\ell}$. If

$$d_{\mathrm{sk}} \ge \frac{8}{\eta^2} \log \frac{4qM\mathcal{M}_k}{\delta}, \quad (41)$$

then, with probability at least $1-\delta$, simultaneously for every such (S, e),

$$\frac{1-\eta}{1+\eta} G_q(e \mid S) \le \widehat{G}(e \mid S) \le \frac{1+\eta}{1-\eta} G_q(e \mid S). \tag{42}$$

In particular, this event covers every trajectory of at most k adaptive selections using these same sketches. It also implies a conventional relative-error bound with $\varepsilon_c = 2\eta/(1-\eta) < 1$.

Proof. For either Gaussian sketch $W \in \{Z, U\}$ and any fixed nonzero vector x of compatible dimension, $d_{\mathrm{sk}}\|W^\top x\|^2/\|x\|^2$ is $\chi^2_{d_{\mathrm{sk}}}$. The moment generating function $\mathbb{E}e^{\theta\chi^2_{d_{\mathrm{sk}}}} = (1-2\theta)^{-d_{\mathrm{sk}}/2}$ and Chernoff optimization give, for $0 < a < 1$,

$$\Pr\{\chi^2_{d_{\mathrm{sk}}} \ge d_{\mathrm{sk}}(1+a)\} \le e^{-d_{\mathrm{sk}}(a-\log(1+a))/2},$$

and $\Pr\{\chi^2_{d_{\mathrm{sk}}} \le d_{\mathrm{sk}}(1-a)\} \le e^{-d_{\mathrm{sk}}(-a-\log(1-a))/2}$. Since $a - \log(1+a) \ge a^2/4$ and $-a - \log(1-a) \ge a^2/2$, the squared-norm failure probability is at most $2e^{-d_{\mathrm{sk}}a^2/8}$. Zero vectors are preserved exactly.

There are at most $2qM\mathcal{M}_k$ deterministic norm queries over all (S, e, j) before the sketches are drawn. A union bound with $a = \eta$ gives failure probability at most $4qM\mathcal{M}_k e^{-d_{\mathrm{sk}}\eta^2/8} \le \delta$. On the complementary event, both ϕ and z are preserved within $1 \pm \eta$. Moreover,

$$z + (\sigma + t_j)\phi = v^\top(\mathsf{A}_S + t_j I)v = v^\top b_e = d.$$

Thus $(1-\eta)(1+d) \le 1 + \widehat{z} + (\sigma+t_j)\widehat{\phi} \le (1+\eta)(1+d)$. Dividing the numerator bounds by these positive denominator bounds and summing with $\omega_j > 0$ proves (42). Its upper factor is $1+\varepsilon_c$ and its lower factor is at least $1-\varepsilon_c$. The event covers all admissible sets simultaneously, hence also every adaptive trajectory. □

Corollary 2 (Global orthogonal-column variant)**.** *In Corollary 1, replace the two Gaussian sketches by independent scaled Haar frames $\sqrt{d_i/d_{\mathrm{sk}}}\,Q_i$, where the two Q_i have d_{sk} orthonormal columns in dimensions $d_1 = n$ and $d_2 = m_0 + M$, respectively. For $d_{\mathrm{sk}} \le \min(d_1, d_2)$, the same conclusion holds under the sufficient condition*

$$d_{\mathrm{sk}} \ge \frac{72}{\eta^2} \log \frac{8qM\mathcal{M}_k}{\delta}. \tag{43}$$

The node sketch may instead be a Haar frame in $\mathbf{1}^\perp$, scaled by $\sqrt{(n-1)/d_{\mathrm{sk}}}$, with $d_1 = n-1$.

Proof. Rotational invariance gives, for a fixed nonzero $x \in \mathbb{R}^d$,

$$\frac{\|\sqrt{d/d_{\mathrm{sk}}}\,Q^\top x\|^2}{\|x\|^2} \overset{\mathrm{d}}{=} \frac{A/d_{\mathrm{sk}}}{(A+B)/d},$$

where $A \sim \chi^2_{d_{\mathrm{sk}}}$ and $B \sim \chi^2_{d-d_{\mathrm{sk}}}$ are independent. Apply the preceding chi-square bound to A and $A+B$ with tolerance $\eta/3$. When both normalized variables lie in $1 \pm \eta/3$, their ratio lies in $1 \pm \eta$, since $2\eta/(3-\eta) \le \eta$ and $2\eta/(3+\eta) \le \eta$. The fixed-vector failure probability is therefore at most $4e^{-d_{\mathrm{sk}}\eta^2/72}$ for $d_{\mathrm{sk}} \le d$. Union bounding the $2qM\mathcal{M}_k$ queries proves the stated dimension bound; the score comparison follows from the same positive-denominator argument.

TABLE I
GUARANTEES AND COVERAGE OF DEFAULT REAL-NETWORK RUNS. "EXACT ARITHMETIC" EXCLUDES SOLVE AND ROUNDOFF ERRORS.

Component	Guarantee	Default-run coverage
Quadrature (Thm. 1)	Deterministic	Yes: prescribed rule, exact arithmetic.
Ratio/curvature (Thm. 2)	Structural	Yes; conservative.
Fixed sketch (Cor. 1)	High probability	No: Gaussian construction and dimension required.
End-to-end (Thm. 3)	Conditional	Total errors not certified.
Maintenance (Thm. 4)	Algebraic/residual	Exact identities; numerical bound conditional.
Heap (Thm. 5)	Exact arithmetic	Lossless proxy ranking; not the GPU backend.
Round recovery (Sec. V-F)	Empirical	Means 0.964–1.000; minima 0.900–1.000.

For centered node sketches, $\mathbf{1}^\top v_{S,e,j} = \mathbf{1}^\top b_e/(\sigma + t_j) = 0$. Every projected node vector therefore lies in $\mathbf{1}^\perp$, where the preceding argument applies with dimension $n-1$. □

a) Guarantees and default-run coverage: Table I distinguishes mathematical guarantees from the evidence supplied by default GPU runs. The GPU composite rule uses $r = 8$ at $\epsilon = 10^{-2}$, with analytic error bound $256\,4^{-8} = 1/256$; prescribed pole counts match $q = 72$–152 in Table II. The controlled experiment's separate fixed $q = 20$ rule is not assigned this certificate.

For $M = 2000$, $k = 20$, $q = 80$, $\eta = 1/4$, and $\delta = 0.05$, Corollary 1 requires $d_{\mathrm{sk}} \ge 15535$; the default uses 128 block-orthogonal rather than Gaussian columns. Computed CG residuals and absolute maintenance bounds do not certify Theorem 3's total relative-error conditions. Even for exact greedy, its exponential bound approaches 0.345 at $\sigma = 1$, $s = 0.5$, and large Λ.

IV. THE POLAR FRAMEWORK

POLAR evaluates G_q using shared shifted solves and maintains $\widehat{G}$ across rounds. Fix a state S and write $\mathsf{A} = \mathsf{A}_S$, $R_j = (\mathsf{A} + t_j I)^{-1}$. Projections Z, U are as in Corollary 1; sketches retain only candidate endpoint rows $V_{\mathcal{C}}$.

A. Shared Krylov Iterations and Full-Norm Tracking

For a common right-hand side w, the shifted systems $(\mathsf{A} + t_j I)u_j = w$ share a Krylov space. Order the poles so that $t_1 \le \cdots \le t_q$, take $A = \mathsf{A} + t_1 I$ as the seed matrix, and let $\theta_j = t_j - t_1$. Standard multi-shift CG shares seed sparse matrix–vector products (SpMVs) across poles, with shifted residuals proportional to the seed residual, $r_i^{(j)} = \zeta_i^{(j)} r_i$, and shifted step coefficients $a_i^{(j)}, \varrho_i^{(j)}$ [8]. Here r_i is the seed residual, and $u_i^{(j)}, d_i^{(j)}$ are the shifted solution and search direction, initialized by $u_0^{(j)} = 0$, $r_0 = d_0^{(j)} = w$, and $\zeta_0^{(j)} = 1$; $w = 0$ returns immediately. A shifted pole freezes when $|\zeta_i^{(j)}|\|r_i\| \le \mathrm{tol}_j$, where tol_j is its residual threshold.

At iteration i, let u_i, r_i, and d_i denote the seed conjugate-gradient (CG) iterate, residual, and search direction. Initialize $u_0 = 0$ and $r_0 = d_0 = w$. Every shifted system starts from $u_0^{(j)} = 0$ and $d_0^{(j)} = w$; $w = 0$ is returned immediately.

With a_i and ϱ_i denoting the seed CG step length and direction coefficient, respectively, one seed iteration is

$$a_i = \frac{r_i^\top r_i}{d_i^\top A d_i}, \qquad r_{i+1} = r_i - a_i A d_i, \tag{44}$$

$$\varrho_i = \frac{\|r_{i+1}\|^2}{\|r_i\|^2}, \qquad d_{i+1} = r_{i+1} + \varrho_i d_i. \tag{45}$$

Only $Ad_i = \mathsf{A}d_i + t_1 d_i$ requires a graph SpMV. For pole j, let $\zeta_i^{(j)}$ be its residual scaling factor and $a_i^{(j)}, \varrho_i^{(j)}$ its shifted coefficients. Initialize $\zeta_{-1}^{(j)} = \zeta_0^{(j)} = 1$, $\chi_0 = 0$, and $\chi_i = a_i \varrho_{i-1}/a_{i-1}$ for $i > 0$. The recurrences [8] are

$$\zeta_{i+1}^{(j)} = \frac{\zeta_i^{(j)}}{1 + a_i\theta_j + \chi_i(1 - \zeta_i^{(j)}/\zeta_{i-1}^{(j)})}, \tag{46}$$

$$a_i^{(j)} = a_i \frac{\zeta_{i+1}^{(j)}}{\zeta_i^{(j)}}, \tag{47}$$

$$\varrho_i^{(j)} = \varrho_i \left(\frac{\zeta_{i+1}^{(j)}}{\zeta_i^{(j)}}\right)^2, \tag{48}$$

$$u_{i+1}^{(j)} = u_i^{(j)} + a_i^{(j)} d_i^{(j)}, \tag{49}$$

$$d_{i+1}^{(j)} = \zeta_{i+1}^{(j)} r_{i+1} + \varrho_i^{(j)} d_i^{(j)}. \tag{50}$$

Here and below, Krylov identities are stated in exact arithmetic. A pole freezes once $|\zeta_{i+1}^{(j)}|\|r_{i+1}\| \le \mathrm{tol}_j$, where tol_j is its prescribed residual threshold; subsequent iterations omit its vector and norm updates.

Re-ranking needs full solution norms even when shifted vectors retain only endpoint entries. The following exact-arithmetic tracker recovers them without full shifted-vector storage.

Proposition 3 (Pole sharing and three-scalar norm tracking)**.** *All shifted systems share the seed SpMVs. Apart from the full-length seed CG vectors, separate shifted solution and search vectors need only be stored on $V_\mathcal{C}$. Their full solution norms are recovered by three scalars per pole, $\mu_i^{(j)} = \|u_i^{(j)}\|^2$, $c_i^{(j)} = u_i^{(j)\top} d_i^{(j)}$, and $v_i^{(j)} = \|d_i^{(j)}\|^2$, initialized by $\mu_0^{(j)} = c_0^{(j)} = 0$ and $v_0^{(j)} = \|w\|^2$. Suppressing j, their updates are*

$$\mu_{i+1} = \mu_i + 2a_i^{(j)} c_i + (a_i^{(j)})^2 v_i, \tag{51}$$

$$c_{i+1} = \varrho_i^{(j)}(c_i + a_i^{(j)} v_i), \tag{52}$$

$$v_{i+1} = (\zeta_{i+1}^{(j)})^2 \|r_{i+1}\|^2 + (\varrho_i^{(j)})^2 v_i. \tag{53}$$

Proof. Binomial expansion gives $\mathcal{K}_i(A + \theta_j I, w) \subseteq \mathcal{K}_i(A, w)$; applying it to $A = (A + \theta_j I) - \theta_j I$ gives the reverse inclusion. Let P_i be the seed residual polynomial, so $r_i = P_i(A)w$ and $P_i(0) = 1$. Its roots are positive Ritz values. Therefore $P_i(-\theta_j) > 0$, and $Q_i(z) = P_i(z-\theta_j)/P_i(-\theta_j)$ has $Q_i(0) = 1$. The residual $Q_i(A+\theta_j I)w$ is a multiple of r_i, hence orthogonal to the shared $\mathcal{K}_i$. Moreover, $(1 - Q_i(z))/z$ is a polynomial of degree at most $i-1$, so the corresponding iterate belongs to that space. Uniqueness of the SPD Galerkin iterate proves $r_i^{(j)} = \zeta_i^{(j)} r_i$ with $\zeta_i^{(j)} = 1/P_i(-\theta_j)$.

Eliminating d_i from seed CG gives $P_{i+1}(z) = (1 + \chi_i - a_i z)P_i(z) - \chi_i P_{i-1}(z)$. Evaluation at $-\theta_j$ gives the stated ζ recurrence. Matching the coefficient of z in the shifted recurrence gives $a_i^{(j)}$; taking the ratio of successive residual norms gives $\varrho_i^{(j)}$. Thus (50) produces the shifted CG iterates without another SpMV. Since $P_i(-\theta_j) = \prod_\ell (1 + \theta_j/\lambda_\ell^{(i)}) \ge 1$, all shifted residuals meet a common tolerance when the seed does.

CG orthogonality gives $r_{i+1}^{(j)} \perp \mathcal{K}_{i+1}$, which contains both $u_{i+1}^{(j)}$ and $d_i^{(j)}$. Expanding the squared norm of $u_i^{(j)} + a_i^{(j)} d_i^{(j)}$ proves the first tracker identity. Taking its inner product with $r_{i+1}^{(j)} + \varrho_i^{(j)} d_i^{(j)}$ proves the second. Expanding the squared norm of this search direction and using $r_{i+1}^{(j)\top} d_i^{(j)} = 0$ proves the third. These scalar recurrences use the full seed residual norm, while every shifted vector recurrence may be restricted to $V_\mathcal{C}$ coordinatewise. □

With at most h seed iterations and m current edges, one right-hand side uses h SpMVs and $O(h(m + n + qn_\mathcal{C}))$ arithmetic before pole-freezing savings; the seed condition number is at most $(\Lambda + t_1)/(\sigma + t_1)$. Finite-precision residual and norm-tracking errors must be included when certifying ε_r in Theorem 3.

B. Cross-Round Endpoint Maintenance

To avoid rebuilding the two sketch banks, let $W_S = C_S^\top U$, $X_j = (\mathsf{A}_S + t_j I)^{-1} Z$, and $Y_j = (\mathsf{A}_S + t_j I)^{-1} W_S$, stored on $V_\mathcal{C}$. For a selected edge with incidence vector b, let $g \in \mathbb{R}^{d_{\mathrm{sk}}}$ denote its candidate row in U. Then W_S changes by $bg^\top$. With $u_j = (\mathsf{A}_S + t_j I)^{-1} b$, $D_j = 1 + b^\top u_j$, $\xi_j = X_j^\top b$, and $\upsilon_j = g - Y_j^\top b$, the same selected-edge solve updates both banks.

Theorem 4 (Slice maintenance and residual accumulation)**.** *The exact updates are*

$$X_j' = X_j - u_j \xi_j^\top / D_j, \qquad Y_j' = Y_j + u_j \upsilon_j^\top / D_j. \tag{54}$$

They can be evaluated solely on candidate endpoint rows in $O(qn_\mathcal{C} d_{\mathrm{sk}})$ time. Fix a pole j. At round r, let b_r be the inserted edge's incidence vector, $A_{r,j} = \mathsf{A}_{S_r} + t_j I$, and $\lambda_{\mathrm{lo,j}} = \sigma + t_j$. Write $X_{r,j}$ for X_j at state S_r, $\widetilde{X}_{r,j}$ for its computed value, and $\|\cdot\|_F$ for the Frobenius norm. Suppose $\widetilde{u}_{r,j}$ approximates $A_{r-1,j}^{-1} b_r$, with residual $f_{r,j} = A_{r-1,j}\widetilde{u}_{r,j} - b_r$ satisfying $\|f_{r,j}\|_2 \le \mathrm{tol}_j$, and define $F_{r,j} = A_{r,j}\widetilde{X}_{r,j} - Z$. Use $\widetilde{D}_{r,j} = 1 + b_r^\top \widetilde{u}_{r,j}$ in the updates. If $k\mathrm{tol}_j \le \sqrt{\lambda_{\mathrm{lo,j}}}$, then

$$\|\widetilde{X}_{k,j} - X_{k,j}\|_F \le \frac{\exp(1)}{\lambda_{\mathrm{lo,j}}}\|F_{0,j}\|_F + \frac{\exp(1) k\mathrm{tol}_j}{\lambda_{\mathrm{lo,j}}^{3/2}}\|Z\|_F. \tag{55}$$

In particular, exact initialization gives the second term alone.

Proof. Write $A = \mathsf{A}_S + t_j I$ and $u = u_j$. Sherman–Morrison gives $(A + bb^\top)^{-1} = A^{-1} - uu^\top/(1 + b^\top u)$. Multiplication by Z proves the first identity in (54). Multiplication by $W_S + bg^\top$, using $u^\top W_S = b^\top Y_j$ and $1 - (b^\top u)/(1 + b^\top u) = 1/(1 + b^\top u)$, proves the second. Restricting these outer products to $V_\mathcal{C}$ is exact because every queried b_e is supported there.

For the residual bound, fix j and put $\vartheta_j = \mathrm{tol}_j/\sqrt{\lambda_{\mathrm{lo,j}}} \le 1$ and $d = b_r^\top A_{r-1,j}^{-1} b_r$. Since $A_{r-1,j} \succeq \lambda_{\mathrm{lo,j}} I$, we have $|b_r^\top A_{r-1,j}^{-1} f_{r,j}| \le \vartheta_j\sqrt{d}$ and $\|b_r^\top A_{r-1,j}^{-1}\|_2 \le \sqrt{d/\lambda_{\mathrm{lo,j}}}$. With $1 + d \ge 2\sqrt{d}$,

$$\widetilde{D}_{r,j} \ge 1 + d - \vartheta_j\sqrt{d} \ge 3/4, \qquad \frac{\|b_r^\top A_{r-1,j}^{-1}\|}{\widetilde{D}_{r,j}} \le \lambda_{\mathrm{lo,j}}^{-1/2}.$$

Because $A_{r,j}\widetilde{u}_{r,j} = b_r\widetilde{D}_{r,j} + f_{r,j}$, multiplying the computed X update by $A_{r,j}$ yields

$$F_{r,j} = F_{r-1,j} - f_{r,j}\frac{b_r^\top \widetilde{X}_{r-1,j}}{\widetilde{D}_{r,j}}.$$

Using $\widetilde{X}_{r-1,j} = A_{r-1,j}^{-1}(Z + F_{r-1,j})$ gives $\|F_{r,j}\|_F \le (1+\vartheta_j)\|F_{r-1,j}\|_F + \vartheta_j\|Z\|_F$. Induction gives $\|F_{k,j}\|_F \le (1+\vartheta_j)^k\|F_{0,j}\|_F + [(1+\vartheta_j)^k - 1]\|Z\|_F$; since $k\vartheta_j \le 1$ implies $(1+\vartheta_j)^k \le \exp(1)$ and $(1+\vartheta_j)^k - 1 \le \exp(1)k\vartheta_j$, this yields $\|F_{k,j}\|_F \le \exp(1)(\|F_{0,j}\|_F + k\vartheta_j\|Z\|_F)$. Finally, $\widetilde{X}_{k,j} - X_{k,j} = A_{k,j}^{-1}F_{k,j}$ and $\|A_{k,j}^{-1}\| \le 1/\lambda_{\mathrm{lo,j}}$ prove (55). □

For $Y_{r,j}$, define $F_{r,j}^Y = A_{r,j}\widetilde{Y}_{r,j} - W_r$, where $W_r = W_{S_r}$. With g_r denoting the inserted edge's projection row, direct multiplication gives

$$F_{r,j}^Y = F_{r-1,j}^Y + f_{r,j}\frac{(g_r - \widetilde{Y}_{r-1,j}^\top b_r)^\top}{\widetilde{D}_{r,j}}.$$

The preceding bounds give $\|F_{r,j}^Y\|_F \le (1+\vartheta_j)\|F_{r-1,j}^Y\|_F + \vartheta_j\|W_{r-1}\|_F + \frac{4}{3}\mathrm{tol}_j\|g_r\|_2$. For $W_{\max} = \max_{0\le r<k}\|W_r\|_F$, iteration yields

$$\|\widetilde{Y}_{k,j} - Y_{k,j}\|_F \le \frac{\exp(1)}{\lambda_{\mathrm{lo,j}}}\Bigg(\|F_{0,j}^Y\|_F + k\vartheta_j W_{\max} + \frac{4\mathrm{tol}_j}{3}\sum_{r=1}^{k}\|g_r\|_2\Bigg).$$

C. Shared-Direction Algebraic Norm Caching

The updates in (54) share directions ξ_j, υ_j across all candidates. Rather than recomputing projected norms at $O(qMd_{\mathrm{sk}})$ cost per round, POLAR maintains their squared values through candidate-dependent scalar coefficients.

Proposition 4 (Shared-direction norm cache)**.** *For every remaining candidate e, let*

$$\phi_{ej} = \|X_j^\top b_e\|^2, \qquad z_{ej} = \|Y_j^\top b_e\|^2, \qquad c_{ej} = b_e^\top u_j/D_j.$$

Compute the two shared directions $w_j^X = X_j\xi_j$ and $w_j^Y = Y_j\upsilon_j$. Then the exact cache updates are

$$\phi_{ej}^+ = \phi_{ej} - 2c_{ej}b_e^\top w_j^X + c_{ej}^2\|\xi_j\|^2, \tag{56}$$

$$z_{ej}^+ = z_{ej} + 2c_{ej}b_e^\top w_j^Y + c_{ej}^2\|\upsilon_j\|^2. \tag{57}$$

Maintaining both sketch slices and all candidate caches costs $O(qn_Cd_{\mathrm{sk}} + qM)$ per selected edge.

Proof. By (54), with $x = X_j^\top b_e$ and $y = Y_j^\top b_e$,

$$x^+ = x - c_{ej}\xi_j, \qquad y^+ = y + c_{ej}\upsilon_j.$$

Expanding these two squared norms gives (56)–(57), since $\langle x, \xi_j\rangle = b_e^\top X_j\xi_j$ and $\langle y, \upsilon_j\rangle = b_e^\top Y_j\upsilon_j$. Only rows in V_C are needed for w_j^X, w_j^Y, and each remaining quadratic-form coefficient is an endpoint difference. Computing the shared directions and applying the slice updates therefore costs $O(qn_Cd_{\mathrm{sk}})$; all candidate scalar recurrences cost $O(qM)$. □

The cache backend reads $O(q)$ scalars per candidate; a lazy backend instead materializes projected norms only when required. The GPU implementation uses cached norms.

D. Lossless Non-Submodular Lazy Selection

With $\lambda_{\mathrm{lo,j}} = \sigma + t_j$, the cached proxy is

$$\widehat{G}(e \mid S) = \sum_j \omega_j\frac{\phi_{ej}}{1 + z_{ej} + \lambda_{\mathrm{lo,j}}\phi_{ej}}. \tag{58}$$

The GPU backend sorts these scores directly. A CPU-oriented lazy alternative bounds proxy changes rather than treating stale scores as upper bounds. For insertion ℓ, let $u_{\ell j}$ and $D_{\ell j}$ be its selected-edge solution and denominator. Then $|c_{e\ell j}| = |b_e^\top u_{\ell j}/D_{\ell j}| \le \chi_{\ell j}$, where

$$\chi_{\ell j} = \frac{\max_{v\in V_C} u_{\ell j}(v) - \min_{v\in V_C} u_{\ell j}(v)}{D_{\ell j}},$$

with $u_{\ell j}(v)$ the entry at vertex v.

Theorem 5 (Common-affine heap bound)**.** *Define the score-change bound and its prefix by*

$$D_\ell^{\mathrm{heap}} = \sum_j \omega_j\chi_{\ell j}\left(\frac{\|\xi_{\ell j}\|}{\sqrt{\lambda_{\mathrm{lo,j}}}} + \frac{\|\upsilon_{\ell j}\|}{4\lambda_{\mathrm{lo,j}}}\right), \qquad P(r) = \sum_{\ell=1}^{r} D_\ell^{\mathrm{heap}}, \quad P(0) = 0. \tag{59}$$

For a candidate last evaluated at r_0, the persistent key $K_e = \widehat{G}(e \mid S_{r_0}) - P(r_0)$ satisfies $\widehat{G}(e \mid S_r) \le K_e + P(r)$ for $r \ge r_0$. This upper bound costs $O(1)$ to query. A max-heap ordered by K_e, with exact current-proxy refinement, recovers the same proxy top-B shortlist as full evaluation under identical lexicographic tie-breaking.

Proof. Put $x = X_j^\top b_e$, $y = Y_j^\top b_e$, and $\psi_j(x,y) = \|x\|^2/(1 + \|y\|^2 + \lambda_{\mathrm{lo,j}}\|x\|^2)$. Direct differentiation gives $\|\nabla_x\psi_j\| \le \lambda_{\mathrm{lo,j}}^{-1/2}$ and $\|\nabla_y\psi_j\| \le (4\lambda_{\mathrm{lo,j}})^{-1}$. Equation (54) changes x, y by $-c_{e\ell j}\xi_{\ell j}, c_{e\ell j}\upsilon_{\ell j}$. The mean-value bound and positive weights therefore give $|\widehat{G}(e \mid S_\ell) - \widehat{G}(e \mid S_{\ell-1})| \le D_\ell^{\mathrm{heap}}$. Summing yields the claimed bound. Since $P(r)$ is common, ordering K_e orders current upper bounds. Refine the largest bound by evaluating its current proxy and resetting its key. Once B evaluated candidates dominate all remaining upper bounds, including tie-breaking, they are the full-evaluation top-B. □

This exact-arithmetic result concerns proxy ranking, not true-marginal ranking or certified floating-point comparisons.

Algorithm 1: POLAR with cached or lazily materialized proxy norms

Input: $\mathsf{A}_0, \mathcal{C}, k, s, \sigma, d_{\mathrm{sk}}, B$; quadrature tolerance ϵ; relative solve tolerance tol; backend cache or lazy
Output: Selected set S_k

Construct positive quadrature (ω_j, t_j) at tolerance ϵ and fixed projections Z, U; set $S = \varnothing$, $\mathsf{A} = \mathsf{A}_0$
Use $2d_{\mathrm{sk}}$ multi-shift solves to initialize endpoint slices X_j, Y_j
Initialize candidate proxy caches or lazy proxy anchors; for heap ranking, set $P(0) = 0$ and $K_e = \widehat{G}(e \mid S_0)$
for $r = 0, \ldots, k-1$ **do**
  Obtain the proxy top-$\min\{B, |\mathcal{C} \setminus S|\}$ shortlist $\mathcal{B}$ by direct sorting of cached scores or common-affine heap refinement (Theorem 5)
  Evaluate G_q for each $e \in \mathcal{B}$ by one residual-controlled multi-shift solve at tolerance tol; select the highest-scoring candidate $e^\star$ and retain its per-pole solutions
  if $r + 1 < k$ **then**
    From the old slices compute ξ_j, υ_j, D_j; update cached norms by (56)–(57) where applicable, then slices by (54), reusing $e^\star$'s solutions
    For heap ranking, reinsert losing shortlisted candidates with $K_e = \widehat{G}(e \mid S_r) - P(r)$ and advance $P(r+1)$ by (59)
  $S \leftarrow S \cup \{e^\star\}$; $\mathsf{A} \leftarrow \mathsf{A} + b_{e^\star} b_{e^\star}^\top$
return S

E. Algebraic Re-ranking with Winner Reuse and Cost

Each proxy top-B candidate requires one multi-shift solve for G_q. The winning solutions also update the sketches by (54), without a separate maintenance solve. Initialization and k rounds therefore use at most $2d_{\mathrm{sk}} + kB$ right-hand sides, versus $2kd_{\mathrm{sk}}$ for sketch rebuilding. At h seed iterations per right-hand side, this costs at most $(2d_{\mathrm{sk}} + kB)h$ seed SpMVs; scoring, pole-dependent arithmetic, and sketch maintenance are additional.

a) Total cost: Let N_{heap} count heap operations, N_{ref} count exact proxy refinements, and T_{proj} denote projection construction cost. The analyzed cache-backed heap has arithmetic bound

$$\begin{aligned} O\big(&T_{\mathrm{proj}} + (2d_{\mathrm{sk}} + kB)h(m_0 + k + n + qn_{\mathcal{C}}) \\ &+ qMd_{\mathrm{sk}} + k(qn_{\mathcal{C}}d_{\mathrm{sk}} + qM) \\ &+ N_{\mathrm{ref}}q + N_{\mathrm{heap}} \log M\big). \end{aligned} \tag{60}$$

Lazy norm materialization replaces $N_{\mathrm{ref}}q$ by $N_{\mathrm{ref}}qd_{\mathrm{sk}}$. GPU direct sorting instead contributes at most $O(kM \log M)$ comparison work in place of the heap terms. Projection construction costs $O((n + m_0 + M)d_{\mathrm{sk}})$ for Gaussian sketches or $O((n + m_0 + M)d_{\mathrm{sk}}^2)$ for Gaussian–QR orthogonalization. Storage is $O((n + m_0 + M)d_{\mathrm{sk}} + qn_{\mathcal{C}}d_{\mathrm{sk}} + qM)$. Unpreconditioned CG requires $h = O(\sqrt{\kappa_*} \log(\kappa_*/\mathrm{tol}))$, where $\kappa_* = \Lambda/\sigma$, under the stated residual criterion.

V. Experimental Evaluation

We evaluate optimization accuracy, external-baseline performance, and reuse savings, then compare objective-dependent edge allocations and diffusion responses. Parameter checks separate practical choices from the sufficient conditions in Table I.

A. Setup

a) Datasets: We evaluate eighteen undirected networks from KONECT [14] and SNAP [15]. Four small networks (Zachary, Dolphins, LesMis, and Polbooks) support exhaustive search, and five medium networks (Hamster, email-Eu-core, ca-GrQc, ca-HepPh, and ca-CondMat) support exact-greedy comparisons. The larger networks appear in Table II, reaching 4.0 million vertices and 34.7 million edges on com-LiveJournal. Graphs are treated as undirected, with duplicate edges and self-loops removed; all vertices are retained, including disconnected components.

b) Candidates and parameters: Unless stated otherwise, experiments use $\sigma = 1$, $k = 20$, and $M = 2000$ uniformly sampled missing edges per graph. A separate 2-hop pool samples missing pairs by random two-step walks; small networks use $M = 40$. POLAR uses $d_{\mathrm{sk}} = 128$ block-orthogonal sketches, shortlist size $B = 16$, quadrature tolerance $\epsilon = 10^{-2}$ ($q = 72$–152), and relative CG tolerance 10^{-7}, with projection seed 2026.

c) Baselines: Exact fractional greedy uses analytic quadrature error intervals ($\epsilon = 10^{-9}$) and dense eigensolvers for surviving contenders. Exact-ER and Exact-Logdet optimize inverse-trace and log-determinant gains; Sketched-ER and Sketched-Logdet use their approximate scores. As an external matrix-function estimator baseline, SLQ greedy uses 20 Rademacher probes and 40 Lanczos steps per trace change [16]. No-Reuse is a cross-round maintenance ablation: it rebuilds both sketch banks every round and selects the proxy maximizer. At $B = 1$, POLAR selects the same proxy maximizer and reuses its solution for maintenance. Degree minimizes the current endpoint-degree sum; Random averages ten seeded draws. Final gains use eigenvalues for $n \le 6000$ and otherwise composite quadrature with analytic error target 10^{-9} and CG relative stopping tolerance 10^{-10}. These are numerical references, not verified bounds on total floating-point error.

Proposition 5 (Twin candidates)**.** *Let u, v be nonadjacent in the current graph G_S and have the same weighted neighborhood. Then $g_s(\{u, v\} \mid S) = x^{-s} - (x+2)^{-s}$ with $x = (\mathsf{A}_S)_{uu}$.*

Proof. Equal weighted neighborhoods make $b = \boldsymbol{e}_u - \boldsymbol{e}_v$ an eigenvector of A_S with eigenvalue x. Adding $bb^\top$ increases that eigenvalue by two and leaves the orthogonal complement unchanged, giving the stated marginal. □

d) Implementation and timing: POLAR, No-Reuse, SLQ, and the sketched baselines run on an NVIDIA RTX 4090 using CuPy 14.2 and CSR SpMM. POLAR uses the cache backend with direct GPU sorting of proxy scores; the theoretical heap is not used in these timings. Exact greedy uses GPU eigensolvers for $n \le 6000$ and CPU LAPACK for ca-HepPh and ca-CondMat. Timings include initialization solves, maintenance, scoring, and re-ranking but exclude graph loading and sketch generation. Table II reports medians of three runs where available, two on web-Google and as-Skitter, and one on com-LiveJournal. No-Reuse has three runs on the five smallest

TABLE II

CROSS-ROUND REUSE ABLATION: RUNTIME AND SPMV COUNTS ON THE ELEVEN LARGEST EVALUATED NETWORKS ($s = 0.5$, $k = 20$, $M = 2000$, $d_{\mathrm{sk}} = 128$, $q = 72$–152). SPEEDUPS COMPARE POLAR AT THE INDICATED SHORTLIST SIZE B AGAINST NO-REUSE. FOR TRUNCATED NO-REUSE RUNS, $> T$ (r) INDICATES MORE THAN T SECONDS ELAPSED WITH r COMPLETED ROUNDS; INEQUALITY-MARKED COUNTS AND SPEEDUPS ARE LOWER BOUNDS. POLAR WITH $B = 1$ MATCHES NO-REUSE'S SELECTIONS IN ALL COMPLETED ROUNDS.

			Time (s)					SpMVs ($\times 10^3$)		
Network	n	m	No-Reuse	$B = 1$	Speedup	$B = 16$	Speedup	No-Reuse	$B = 1$	$B = 16$
ca-CondMat	23 133	93 439	29.7	3.3	8.9×	3.8	7.8×	524	27.7	53.1
ca-HepPh	12 008	118 489	43.7	4.2	10.4×	5.6	7.8×	840	43.8	77.5
email-Enron	36 692	183 831	90.6	8.1	11.2×	10.7	8.5×	1236	65.0	121
com-Amazon	334 863	925 872	197.1	12.1	16.3×	20.1	9.8×	622	33.0	62.1
com-DBLP	317 080	1 049 866	193.7	12.2	15.9×	20.0	9.7×	647	34.5	66.6
roadNet-PA	1 088 092	1 541 898	139.6	8.0	17.4×	17.9	7.8×	131	7.0	13.6
roadNet-CA	1 965 206	2 766 607	268.7	15.1	17.8×	32.8	8.2×	136	7.2	13.8
com-Youtube	1 134 890	2 987 624	>3687 (19)	214.8	> 17.2×	542.6	> 6.8×	>3356	187	357
web-Google	875 713	4 322 051	2225.1	125.0	17.8×	303.0	7.3×	2776	147	278
as-Skitter	1 696 415	11 095 298	>4077 (8)	569.9	> 7.2×	1389.5	> 2.9×	>2619	348	656
com-LiveJournal	3 997 962	34 681 189	>2515 (3)	908.8	> 2.8×	1720.9	> 1.5×	>469	166	311

tabulated networks and one elsewhere, with a one-hour limit checked between rounds. Raw results are available in [17].

B. Efficiency

The external SLQ greedy baseline takes 12–39× as long as POLAR on Hamster, email-Eu-core, and ca-GrQc across the three tested exponents, with lower fractional target gains (Table III). This comparison concerns the stated 20-probe, 40-step implementation, not every SLQ configuration. On com-LiveJournal (4.0 million vertices, 34.7 million edges), POLAR finishes in 15.1 minutes at $B = 1$ and 28.7 minutes at the default $B = 16$.

Table II separately isolates cross-round reuse against No-Reuse. At $k = 20$, $d_{\mathrm{sk}} = 128$, rebuilding requires 5120 multi-shift right-hand sides, versus 276 at $B = 1$ and 576 at $B = 16$. On the eight networks where No-Reuse completes, $B = 1$ selects identical edges, uses 18.7–19.2× fewer SpMVs, and runs 8.9–17.8× faster. These are ablation results, not speedups over an external optimizer. Increasing B to 16 costs approximately 1.2–2.5× more time for broader re-ranking. No-Reuse is truncated on com-Youtube, as-Skitter, and com-LiveJournal; inequality-marked speedups are lower bounds. Initialization remains significant: its $2d_{\mathrm{sk}}$ solves account for 94% of completed $B = 1$ SpMVs on web-Google.

C. Effectiveness

All selection-quality comparisons evaluate the resulting edge sets on the specified fractional target f_s, irrespective of the objective used to select them.

a) Comparison with the optimum: Exhaustive search over the four small graphs, $k = 1, \ldots, 5$, and $s \in \{0.25, 0.5\}$ covers forty instances, with up to 658 008 candidate sets per instance. POLAR and exact fractional greedy attain the exhaustive optimum within 2×10^{-11} relative difference throughout. Exact-ER reaches at least 0.9431 of the optimum, Degree at least 0.8026, and Random averages 0.44–0.73. Thus greedy performs well on these tested cases despite the general non-submodularity of f_s.

b) Comparison with exact greedy: Table III normalizes gains by exact fractional greedy for the same s, not by the global optimum. Across the four tabulated medium networks, POLAR achieves 0.9920–1.0000 on uniform pools and 0.9974–1.0000 on 2-hop pools. SLQ attains 0.9327–0.9949 where reported; Section V-B compares runtime. Exact greedy remains competitive on Hamster and email-Eu-core (1–2 s). On ca-HepPh, CPU LAPACK exact greedy takes 1059–1096 s versus 5.6 s for GPU POLAR, a comparison of different implementations rather than hardware-matched speedups. For ca-CondMat, the available $s = 0.5$ exact-greedy reference takes 6471 s; Exact-ER takes 6575 s, with POLAR and Exact-ER achieving $f_{0.5}$ ratios of 0.9944 and 0.9940, respectively.

Competing objectives sometimes perform nearly as well on f_s: on 2-hop pools, Exact-ER and Exact-Logdet lose at most 3.3% and 5.1% relative to exact fractional greedy, while Degree loses up to 17.4% (19.5% on uniform pools). Conversely, Exact-ER or Exact-Logdet exceeds POLAR on f_s in seven of twelve uniform-pool cases (by at most 0.8%) and two of twelve 2-hop cases (by at most 0.03%). In those cases POLAR also trails exact fractional greedy; increasing B to 32 closes the ca-GrQc gap at $s = 0.5$.

c) Large networks: Table IV instead normalizes each f_s gain by the best observed method, without an exact-greedy or optimal reference. POLAR achieves 0.9979–1.0000 of this reference in sixteen of seventeen comparisons. On com-Youtube at $s = 0.5$, default POLAR falls 1.4% below Sketched-ER; raising B to 64 produces 1.0040 times its gain in 1072 s, while $d_{\mathrm{sk}} = 256$ produces 0.9948 in 753 s. The improvement with $B = 64$ indicates sensitivity to the proxy shortlist.

d) Structure of selected edges: The objectives do not choose structurally different edges on Hamster, email-Eu-core, and ca-GrQc: all three fractional POLAR settings and Exact-ER have identical coarse degree summaries within each pool. Medians over the 40 selected endpoint occurrences are $1/0/1$ on uniform pools and $1/2/1$ on 2-hop pools. Fractions of the 20 selected edges touching a base-degree-1-or-2 vertex are $100/50/100\%$ and $100/100/100\%$, respectively.

TABLE III

TARGET f_s GAIN NORMALIZED BY EXACT FRACTIONAL GREEDY AT THE SAME EXPONENT ($k = 20$, $M = 2000$, $\sigma = 1$). EXACT-ER AND EXACT-LOGDET OPTIMIZE DIFFERENT OBJECTIVES, BUT ALL GAINS ARE EVALUATED ON f_s. RANDOM AVERAGES TEN DRAWS; DASHES INDICATE UNREPORTED RESULTS. ON THE CA-GRQC 2-HOP POOL, TWIN CANDIDATES (PROPOSITION 5) PRODUCE UP TO 77 TIED MARGINAL MAXIMIZERS IN A ROUND, AND ALL REPORTED NON-RANDOM METHODS ATTAIN THE SAME GAIN.

		Uniform pool							2-hop pool			
Network	s	POLAR	SLQ	Exact-ER	Exact-Logdet	Sketched-ER	Degree	Random	POLAR	Exact-ER	Exact-Logdet	Degree
email-Eu-core	0.25	0.9949	0.9630	0.9854	0.9942	0.9680	0.9207	0.245	0.9974	0.9669	0.9907	0.9174
	0.5	0.9954	0.9693	0.9967	0.9870	0.9854	0.8731	0.213	0.9989	0.9850	0.9708	0.8724
	0.75	0.9975	0.9949	1.0000	0.9753	0.9943	0.8213	0.189	0.9998	0.9973	0.9489	0.8264
Hamster	0.25	0.9964	0.9327	0.9965	1.0000	0.9965	0.8890	0.480	0.9998	0.9680	1.0000	0.9923
	0.5	1.0000	0.9541	0.9981	1.0000	0.9981	0.8497	0.434	0.9997	0.9819	1.0000	0.9899
	0.75	1.0000	0.9458	0.9997	0.9994	0.9997	0.8060	0.396	1.0000	0.9961	0.9977	0.9851
ca-GrQc	0.25	0.9920	0.9640	0.9944	1.0000	0.9914	0.8838	0.543	1.0000	1.0000	1.0000	1.0000
	0.5	0.9952	0.9541	0.9970	1.0000	0.9944	0.8456	0.500	1.0000	1.0000	1.0000	1.0000
	0.75	0.9981	0.9477	1.0000	0.9997	0.9981	0.8051	0.465	1.0000	1.0000	1.0000	1.0000
ca-HepPh	0.25	0.9942	–	0.9905	0.9973	0.9874	0.9300	0.441	1.0000	0.9754	1.0000	0.9963
	0.5	1.0000	–	0.9954	0.9871	0.9919	0.8927	0.401	1.0000	0.9852	1.0000	0.9952
	0.75	0.9997	–	0.9985	0.9714	0.9948	0.8476	0.367	1.0000	0.9963	0.9985	0.9927

TABLE IV

TARGET f_s GAIN ON SEVEN LARGER NETWORKS, NORMALIZED BY THE LARGEST OBSERVED GAIN AMONG THE REPORTED METHODS FOR EACH TARGET EXPONENT, NOT BY A GLOBAL OPTIMUM. "OTHER s" DENOTES THE BEST POLAR SELECTION OBTAINED AT A DIFFERENT EXPONENT AND EVALUATED ON THE CURRENT TARGET f_s. "SK." ABBREVIATES SKETCHED; DASHES INDICATE UNREPORTED CONFIGURATIONS.

Network	s	POLAR	Other s	Sk.-ER	Sk.-Logdet	Degree	Random
email-Enron	0.25	1.0000	0.9995	0.9935	0.9919	0.8706	0.570
	0.5	1.0000	0.9997	0.9980	0.9862	0.8229	0.525
	0.75	0.9988	0.9970	1.0000	0.9762	0.7677	0.485
com-DBLP	0.25	0.9990	1.0000	0.9819	0.9993	0.9911	0.555
	0.5	1.0000	0.9997	0.9910	0.9946	0.9824	0.517
	0.75	1.0000	0.9922	0.9959	0.9807	0.9641	0.483
com-Amazon	0.25	0.9979	0.9997	0.9821	1.0000	0.9798	0.512
	0.5	1.0000	0.9986	0.9886	0.9978	0.9692	0.470
	0.75	1.0000	0.9973	0.9946	0.9919	0.9536	0.438
roadNet-PA	0.25	0.9981	1.0000	0.9997	0.9987	0.9898	0.646
	0.5	0.9987	1.0000	0.9995	0.9980	0.9857	0.613
	0.75	1.0000	0.9983	0.9993	0.9970	0.9809	0.588
roadNet-CA	0.25	1.0000	1.0000	0.9756	1.0000	0.9885	0.650
	0.5	1.0000	1.0000	0.9833	1.0000	0.9840	0.617
	0.75	1.0000	0.9985	0.9934	0.9984	0.9773	0.591
web-Google	0.5	1.0000	–	0.9874	0.9914	0.9724	0.448
com-Youtube	0.5	0.9858	–	1.0000	0.9766	0.8728	0.658

D. Controlled Multiscale Benchmark

To isolate competition between local and multiscale repairs, we use the controlled graph in Fig. 1(b), comprising two 390-vertex cores built from weight-10 Hamilton cycles, a weight-0.5 bridge, twenty weight-0.3 leaves, and a five-vertex cycle with five weight-0.5 core ties. With $n = 805$, $\sigma = 10^{-2}$, $k = 15$, and a common pool of 2995 intercore or leaf-to-core candidates plus five cycle chords, all methods use graph seeds 711–713.

Table V shows that Exact-ER allocates seven edges between cores, while POLAR selects fifteen leaf repairs at $s = 0.25$ and fourteen leaves plus one intercore edge at $s = 0.5$, matching exact fractional greedy's allocations. Degree instead spends five edges on local cycle chords, whose low endpoint-degree sums favor their selection despite five existing core ties. Under the specified f_s, POLAR exceeds Exact-ER by 48.06% and 21.77% at the two exponents, reaching 99.95% and 99.92% of exact fractional greedy. In the $n = 800$ control without the local cycle, Degree and POLAR are nearly tied at $s = 0.25$ (6.077892 versus 6.078605). The separation is strongest when local and multiscale repair options compete.

A controlled-setting size check increased only (d_{sk}, B) to $(128, 16)$. All three graph seeds retained allocations $0/15/0$ at $s = 0.25$ and $1/14/0$ at $s = 0.5$. Global column-orthogonal sketches and the fixed 20-pole rule were unchanged; this is not a full GPU-default configuration check.

E. Diffusion Time Scales

Figure 2 measures heat-trace reductions at individual diffusion times, separately from integrated fractional gains. At $\sigma = 1$, the Gamma weighting in Proposition 1 has mean $s/\sigma \leq 1$, so similar fractional gains may coexist with different longer-time responses.

On ca-HepPh at $\tau \approx 3.2$, Exact-ER gives the largest reduction among the compared selections; Degree, Exact-Logdet, and POLAR at $s = 0.25, 0.5, 0.75$ achieve 38.2%, 76.3%, and 86.2%, 94.8%, 97.2% of that observed maximum, respectively. Across the evaluated uniform pools, Degree leads at the shortest measured times and inverse-trace selection at the longest, with fractional selections ordered by s in these tests. On 2-hop pools, ratios for $\tau \geq 30$ become rounding-sensitive as the reductions approach zero.

Table VI extends the comparison to three networks using their saved heat curves. At $\tau = 0.1$, Degree leads on Hamster and email-Eu-core, while Exact-Logdet leads on ca-GrQc (the top three entries round to 100%). At $\tau = 10$, Exact-ER leads on Hamster and POLAR at $s = 0.75$ leads on the other two. The preferred observed selection depends on both graph and horizon.

F. Parameters and Robustness

Table VII studies $d_{\mathrm{sk}}, B, \epsilon$ on ca-GrQc at $s = 0.5$. Increasing B to 32 matches exact fractional greedy, while larger d_{sk}

TABLE V

FRACTIONAL TARGET GAINS AND BUDGET ALLOCATIONS ON THE CONTROLLED MULTISCALE GRAPH ($n = 805$, $\sigma = 10^{-2}$, $M = 3000$, $k = 15$). GAINS ARE MEANS $\pm$ SAMPLE STANDARD DEVIATIONS OVER GRAPH SEEDS 711–713. POLAR USES COLUMN-ORTHOGONAL SKETCHES WITH $d_{\mathrm{sk}} = 24$, $B = 4$, AND $q = 20$. Δ_{ER} IS THE PERCENTAGE DIFFERENCE IN TARGET f_s GAIN RELATIVE TO EXACT-ER. C/L/D DENOTES INTERCORE LINKS, LEAF REPAIRS, AND LOCAL CYCLE CHORDS. ALLOCATIONS ARE IDENTICAL ACROSS SEEDS EXCEPT FOR RANDOM, WHOSE MEAN ALLOCATION IS REPORTED.

	$s = 0.25$			$s = 0.5$		
Method	Target f_s gain	Δ_{ER} (%)	C/L/D	Target f_s gain	Δ_{ER} (%)	C/L/D
Random	2.941226 ± 0.586256	-28.36	10/5/0	9.257625 ± 1.170745	-20.83	10/5/0
Degree	4.793361 ± 0.000284	$+16.75$	0/10/5	10.398292 ± 0.000542	-11.08	0/10/5
Exact-ER ($s = 1$)	4.105538 ± 0.000227	$+0.00$	7/8/0	11.694059 ± 0.000555	$+0.00$	7/8/0
Sketched-ER ($d_{\mathrm{sk}} = 24$)	5.094869 ± 0.000657	$+24.10$	4/11/0	13.477414 ± 0.001683	$+15.25$	4/11/0
POLAR	6.078574 ± 0.000183	$+48.06$	0/15/0	14.239384 ± 0.000217	$+21.77$	1/14/0
Exact fractional greedy	6.081501 ± 0.000393	$+48.13$	0/15/0	14.250901 ± 0.000144	$+21.86$	1/14/0

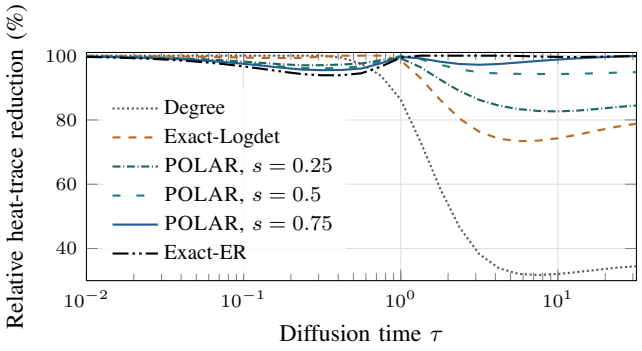


Fig. 2. Heat-trace reduction on ca-HepPh (uniform candidate pool, $k = 20$, $\sigma = 1$), normalized at each diffusion time by the largest reduction among the compared selections, rather than the global optimum. POLAR at $s = 0.5$ selects the same edge set as exact fractional greedy at that exponent. The vertical axis starts at 30%.

TABLE VI

HEAT-TRACE REDUCTION (%) FROM SAVED UNIFORM-POOL SELECTIONS ($\sigma = 1$, $M = 2000$, $k = 20$, $d_{\mathrm{sk}} = 128$, $B = 16$). EACH ROW IS NORMALIZED BY THE BEST OF DEGREE, EXACT-LOGDET, EXACT-ER, AND THE THREE POLAR SETTINGS, NOT BY A GLOBAL OPTIMUM. THE DEGREE/EXACT-LOGDET WINNERS ARE IDENTIFIED IN THE TEXT.

Network	τ	POLAR exponent s			Exact-ER
		.25	.5	.75	
Hamster	0.1	99.94	99.94	99.71	99.50
	10	96.04	96.38	97.92	100.00
email-Eu-core	0.1	90.53	88.03	81.64	81.65
	10	99.99	99.99	100.00	99.75
ca-GrQc	0.1	100.00	100.00	99.10	99.33
	10	94.06	95.02	100.00	99.25

generally improves recovery but not monotonically. Tightening ϵ from 10^{-1} to 10^{-4} leaves gains unchanged to four decimals while increasing q from 60 to 110.

a) Sketch construction: We test sensitivity to the sketch distribution and size. Across five seeds on three medium networks, block-orthogonal sketches at $d_{\mathrm{sk}} = 128, B = 16$ achieve mean ratios of 0.9938–0.9997, versus 0.9887–0.9997 for Gaussian sketches. Reducing to $d_{\mathrm{sk}} = 24, B = 4$ lowers block-orthogonal recovery to 0.9480–0.9909, although the same small configuration is stable on the controlled multiscale benchmark.

b) Per-round recovery and error attribution: We estimate per-round recovery r_i (the selected marginal divided by the largest available marginal) using dense eigendecomposition and quadrature with relative target 10^{-9} to score all candidates along each recorded trajectory. With default parameters, trajectory means are 0.964–1.000 and minima are 0.900–1.000; 58–96% of rounds attain the reference maximum within 10^{-8} relative tolerance. These statistics are neither final-gain ratios nor the certificates of Lemma 4 (Remark 1 in Section III-F).

To locate selection loss, we compare the selected and shortlist-best reference marginals, each normalized by the global reference maximum. Across Hamster, ca-GrQc, and email-Eu-core, we test block-orthogonal sketches at $(d_{\mathrm{sk}}, B) = (24, 4), (128, 16)$ and Gaussian sketches at $(128, 16)$, with five seeds and 20 rounds each: 900 decisions in total. The two normalized scores differ by at most 1.8×10^{-15}. Thus the observed per-round loss arises from shortlist omissions, not from re-ranking within the shortlist. This diagnosis is restricted to the recorded trajectories.

c) Candidate pools, budget, and regularization: These checks vary the default workload. Across three independently sampled uniform pools on each of three medium networks, POLAR recovers 0.9889–1.0000 of exact greedy. With $k = 100$ on the ca-GrQc 2-hop pool, its $s = 0.5$ ratio is 0.9929, versus 0.9793 for Exact-ER, 0.9918 for Degree, and 0.9988 for Exact-Logdet. On the default $k = 20$ uniform pools, POLAR matches exact fractional greedy at $s = 0.5$ for $\sigma \in \{0.01, 0.1\}$; at $\sigma = 0.01$, its Jaccard similarity to Exact-ER is 0.74–1.00, while Degree attains only 0.20–0.58 of the best reported $f_{0.5}$ gain.

TABLE VII

ONE-AT-A-TIME PARAMETER SENSITIVITY ON CA-GRQC WITH UNIFORM CANDIDATES ($s = 0.5$, $\sigma = 1$, $k = 20$, $M = 2000$). RATIO DENOTES THE FINAL f_s GAIN RELATIVE TO EXACT FRACTIONAL GREEDY. UNLESS VARIED, $d_{\mathrm{sk}} = 128$, $B = 16$, AND $\epsilon = 10^{-2}$; POLE COUNT q IS REPORTED FOR THE TOLERANCE SWEEP.

d_{sk}	16	32	64	96	128	192	256
Ratio	0.9694	0.9801	0.9761	0.9934	0.9952	0.9999	0.9987
Time (s)	1.64	1.66	1.83	1.95	2.17	2.48	2.64
B	1	2	4	8	16	32	64
Ratio	0.9796	0.9809	0.9826	0.9952	0.9952	1.0000	1.0000
Time (s)	1.35	1.79	1.99	2.15	2.17	2.23	3.33
ϵ	10^{-1}	10^{-2}	10^{-3}	10^{-4}			
q	60	80	90	110			
Ratio	0.9952	0.9952	0.9952	0.9952			
Time (s)	2.02	2.17	2.24	2.44			

d) Limitations: We measure spectral gains, selected-edge structure, and heat-trace responses, not downstream benefits in deployed graph-management applications. Principal runtime measurements fix $\sigma = 1$, $M = 2000$, and $k = 20$; the quality checks at smaller σ or larger k do not establish runtime scaling in those regimes. Smaller σ worsens the unpreconditioned CG bound. Preconditioning or SDD solve backends [18]–[20], with shift sharing accounted for, remain future work. The common-affine heap has no performance evaluation.

VI. Related Work

a) Spectral objectives and graph modification: Effective-resistance design and discrete edge augmentation prioritize global connectivity [4], [5], while current-flow group measures [1], [2], forest-matrix proximity [3], [21], opinion optimization [22], and Kemeny's constant [23], [24] motivate other spectral quantities. These tasks differ in what an added edge should improve, even when they use related Laplacian representations. FastFrac [9] optimizes a submodular positive-power trace; POLAR optimizes a negative-power trace reduction. Their exponent directions also differ: increasing s flattens the relative positive-power mode weights $(1 + \alpha_{\mathrm{FF}}\lambda)^{s-1}$ but steepens the negative-power weights $(\sigma + \lambda)^{-s-1}$ toward low frequencies. Neither their objectives nor their diminishing-returns properties are interchangeable.

Experimental design studies related spectral criteria through approximate supermodularity [25] and proportional volume sampling for A-optimal design [26]. These address measurement selection rather than fractional graph-edge marginal reuse. Arrigo and Benzi [27] instead optimize network communicability based on adjacency-matrix exponentials.

b) Marginal estimation and computational reuse: Contour quadrature [28] and rational Krylov methods [29] evaluate matrix functions; SLQ estimates traces [16], with numerical analyses in [30], [31]. Edge selection needs relative accuracy for small rank-one trace changes, which separate trace estimates need not provide. Shifted Krylov methods [8], [32] share iterations across poles; SDD solvers [18], [19], sparse elimination [20], and sparsification [12], [33] provide complementary primitives. General low-rank matrix-function updates [34] and rational Krylov extensions [35] already enable reuse. POLAR specializes to fractional edge marginals, sharing shifted solves and maintained quantities across many candidates and rounds.

c) Greedy selection and fixed sketches: Classical lazy greedy relies on diminishing returns [36]; ratio–curvature analysis [7] handles non-submodular objectives. We specialize the latter to fractional gains and combine it with numerical errors. Fixed-sketch projection bounds [13] and lossless proxy ranking address distinct accuracy requirements.

VII. Conclusion

POLAR combines relative-accuracy quadrature, shared solves, and cross-round reuse for fractional inverse-trace selection. Conditional guarantees and experiments distinguish similar-value, different-edge regimes from controlled competition with larger value gaps. Separate heat responses show that different diffusion horizons favor different selections.

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